Faculty of Engineering, Built Environment and Information Technology | Department of IT

Module: ITC327W - Information Technology Work Integrated Learning (2026)

Software Requirements Specification (SRS)

Mangojane Integrated Funeral Management System Prototype

Project Group: The Extended Group (7 Members)
External Stakeholder: Mangojane Funeral Parlor, Welkom
Key Stakeholder Contact: Lerato Moipone Mongojane(Co-Founder and Manager)
Approval Status: Requirements Confirmed & Signed On-Site
Compiled For: Mr. Mokoduwe (ITC327W Module Coordinator) | Unit 1 Progress Checkpoint
Technology Stack: Flutter (Mobile) | ASP.NET Core (Web) | Supabase (PostgreSQL Backend)

Assigned Group Members & Work Allocations (The Extended Group):

Student Name	Student Number	Assigned Project Role	Unit 1 Responsibility
Hlompho Compact Malejane	222025549	Lead Developer / Architect	SRS compilation, Architecture & GitHub Setup
Lehlohonolo Sulvation Modise	222021816	Backend Engineer (Supabase)	Supabase Setup & Data Flow Models
Katleho Daniel Mokhele	221037680	Mobile Developer (Flutter)	Mobile Functional Requirements & Low-Data Specs
Kamohelo Mohlohlo	223010767	Web Developer (ASP.NET Core)	Admin Web Dashboard, Booking Queue & Logistics Dispatch Specs
Nomsa Nomankonya	216006365	Systems & Integration Analyst	Current-Process Flowcharting & DFD Verification
Lesego Precious Waeza	222041368	Requirements Engineer / Lead	Stakeholder Interview & Needs Assessment
Nthabiseng Patricia Moeketsi	211146692	QA & Documentation Specialist	MS Project Baseline & Review Verification

1. PROJECT TITLE

Mangojane Integrated Funeral Management System Prototype

2. PROBLEM STATEMENT

Mangojane Funeral Parlor, situated in Welkom in the Free State Province, operates as a critical community funeral undertaker and burial society administrator. The parlor currently relies entirely on physical paper diaries, manual receipt books, walk-in visits, and telephonic inquiries to manage client service bookings, verify burial society policy records, and dispatch transport assets (hearses, family cars, tents, and lowering equipment).

This manual workflow results in frequent double-booking of mortuary logistics, misplaced policy payment cards, lengthy queue delays for grieving families during urgent bereavement periods, and a lack of real-time operational visibility for parlor dispatchers. A centralized, integrated digital prototype is required to streamline client bookings, automate asset scheduling, and secure policy records across mobile and web platforms.

3. PURPOSE, AIM AND OBJECTIVES

3.1 Purpose

The purpose of this software prototype is to digitize and centralize funeral intake workflows and logistical coordination. It enables bereaved families and policyholders to browse service packages and request funeral arrangements remotely, while providing parlor administrators with a real-time web console to schedule assets, eliminate diary booking clashes, and log premium payments.

3.2 Project Aim

To optimize administrative intake and dispatch efficiency at Mangojane Funeral Parlor through an integrated cross-platform mobile and web software prototype supported by a shared cloud backend.

3.3 Main Objectives

- Provide an accessible, low-bandwidth Flutter mobile application for clients to view funeral service packages, caskets, and submit arrangement booking requests remotely.
- Implement an administrative ASP.NET Core web portal for parlor staff to approve/reject bookings, manage vehicle/staff dispatch calendars, and track burial society premium records.
- Deploy a shared Supabase PostgreSQL cloud backend with Role-Based Access Control (RBAC) and Row-Level Security (RLS) to ensure secure bidirectional data synchronization.
- Execute manual testing with synthetic test datasets to verify cross-system workflows and ensure strict POPIA privacy compliance.

4. PROJECT SCOPE & BOUNDARIES

4.1 Project Deliverables

- **Flutter Mobile Client:** Cross-platform app for clients with package catalog viewer, booking request forms, and real-time status tracker cards.
- **ASP.NET Core Web Portal:** Central management dashboard for parlor administrators to process bookings, assign hearses, and log policy premiums.
- **Supabase Cloud Backend:** Managed PostgreSQL database with Realtime synchronization, JWT authentication, and secure RLS policies
- **Management Artefacts:** Software Requirements Specification (SRS), MS Project Schedule (.mpp), GitHub repository, and Excel Progress Tracker.

4.2 Acceptance Criteria

- Booking requests submitted via Flutter must appear in the ASP.NET admin queue within 3.0 seconds under standard connectivity.
- The system must prevent conflicting hearse/driver allocations for the same burial time-slot on the administrative calendar.
- Data access must be strictly isolated using Supabase RLS so clients can only view their own policy and booking records.

4.3 Exclusions, Constraints and Assumptions

Project Constraints & Assumptions

- **Scope Exclusions:** Live merchant banking / credit card payment gateway integration (simulated via mock payment reference numbers); direct automated DHA population register synchronization; hardware GPS vehicle tracking units.
- **Technical Constraints:** Single academic semester development timeline (ITC327W); mandatory tech stack restricted to Flutter, ASP.NET Core, and Supabase.
- **Assumptions:** Clients have access to entry-level Android smartphones with basic mobile data; parlor office has stable broadband internet connectivity

5. STAKEHOLDERS AND USER GROUPS

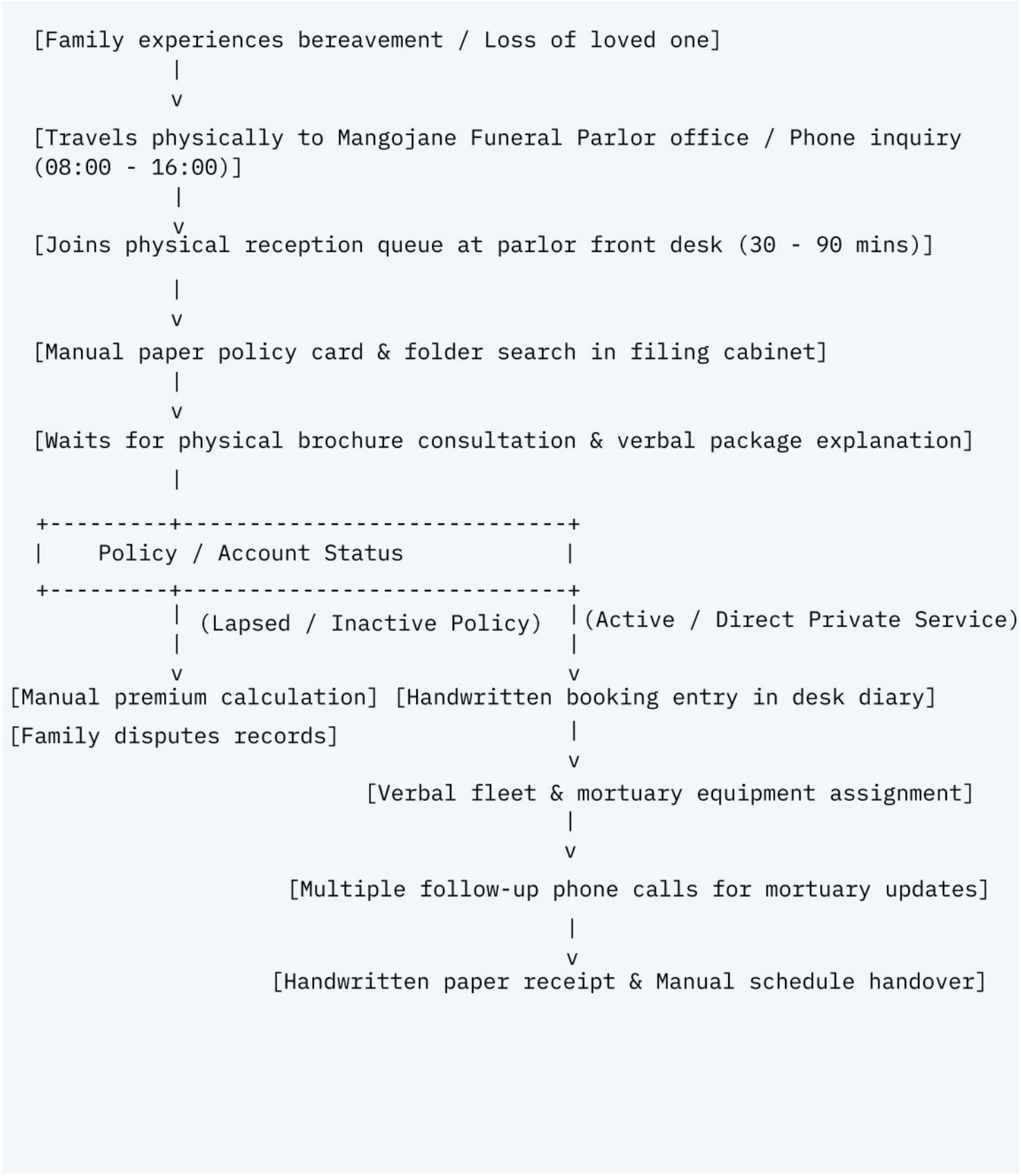
User Group / Stakeholder	Primary Roles & Responsibilities	Interface Used	Core Needs & Constraints
Bereaved Client / Policyholder	Registers profile, browses funeral packages, submits booking requests, and tracks arrangement milestones	Flutter Mobile App	Simple high-contrast UI, low bandwidth consumption, transparent pricing visibility.
Parlor Administrator / Front Desk	Reviews incoming booking queue, verifies policy active status, approves dates, and allocates hearses/equipment.	ASP.NET Web Portal	Centralized booking calendar, conflict free resource scheduling, quick premium payment logging.
Mortuary & Fleet Dispatch Staff	Views daily collection and burial service schedules, vehicle allocations, and cemetery destination routes.	ASP.NET Web Portal	Clear operational schedule summaries, accurate deceased collection notes.
Parlor Management / Owner	Monitors daily service volume, audits policy accounts, reviews operational KPIs, and ensures data privacy	ASP.NET Web Portal	Aggregated monthly reports, revenue summary logs, POPIA compliance oversight

6. CURRENT PROCESS AND PROCESS FLOW DIAGRAM

6.1 Current Process Narrative (Manual Walk-In Workflow)

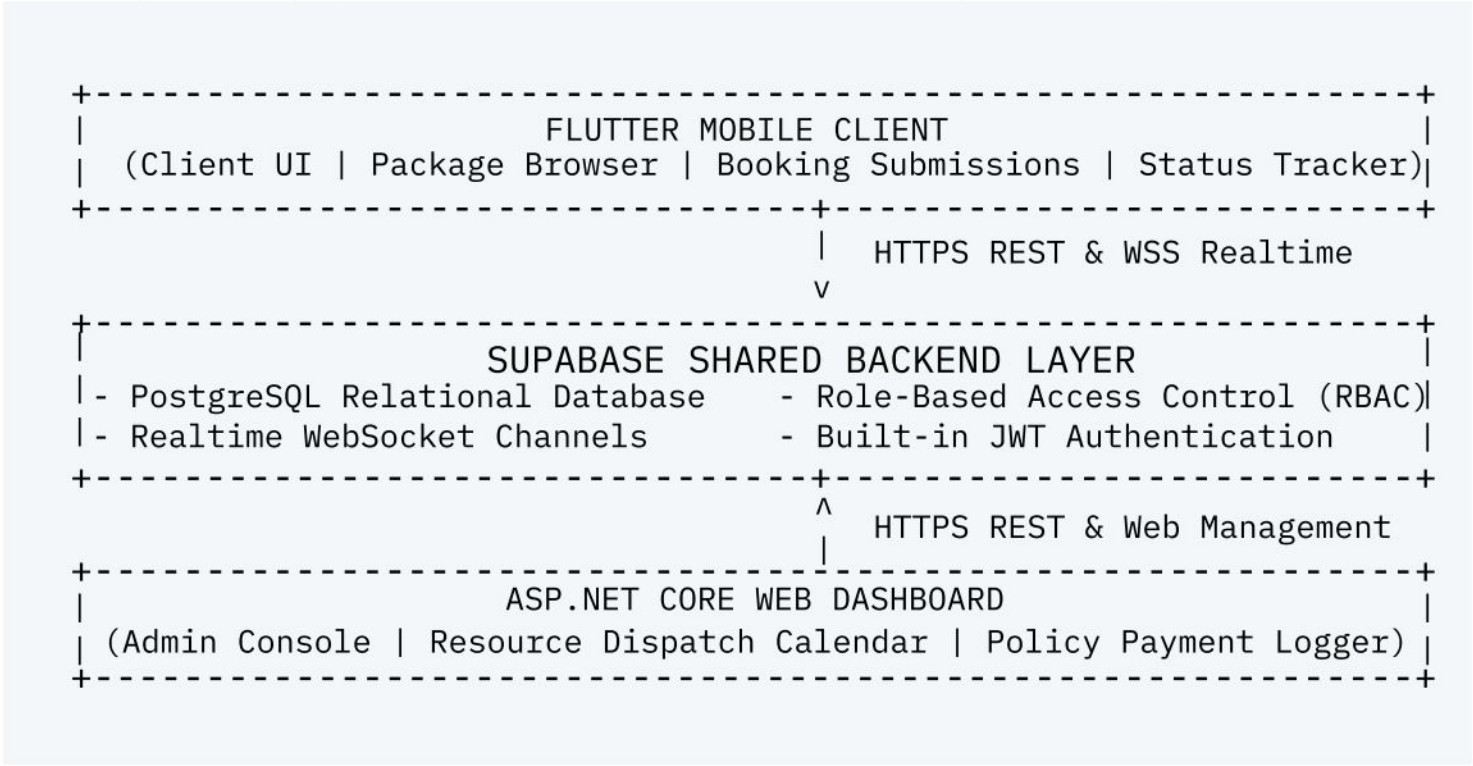
Currently, bereaved family members travel in person to Mangojane Funeral Parlor or phone the office during business hours. Front-desk staff manually locate physical paper folders or burial society cards from filing cabinets. Funeral package options and casket selections are explained verbally using printed brochures. Service bookings, burial dates, hearse allocations, and equipment requirements are manually recorded into a physical desk diary. This manual method frequently causes overlapping vehicle bookings during busy weekends, misplaced premium receipts, and requires families to make multiple follow-up phone calls to check the status of mortuary preparations.

6.1 Current Process Narrative (Manual Walk-In Workflow)



7. PRODUCT PERSPECTIVE

The system operates as a unified, three-tier distributed software prototype designed to decouple client service requests from physical front-desk office visits while centralizing back-office logistics.



8. MAIN PRODUCT FUNCTIONS

System Component	Main Functional Capabilities
Flutter Mobile App	• Client account registration and secure JWT-based email/password authentication. • Funeral package catalog browser with casket specifications and add-on pricing. • Digital funeral booking intake form with deceased details and preferred dates. • Real-time booking status tracker card displaying arrangement milestones (Pending, Approved, Logistics Assigned)
ASP.NET Core Web App	• Administrative booking queue management with live request approval, modification, and rejection controls. • Fleet and mortuary resource dispatch calendar preventing overlapping hearse/equipment bookings. • Burial society policy search, member status verification, and premium payment transaction logger. • Generation of daily dispatch schedule reports for mortuary and transport staff.
Supabase Shared Backend	• Relational PostgreSQL database for storing user accounts, service packages, bookings, and policies. • Realtime change broadcasting via PostgreSQL replication to update mobile and web frontends. • Row-Level Security (RLS) policies enforcing POPIA-compliant data isolation between users.

9. SOFTWARE REQUIREMENTS SPECIFICATIONS

9.1 Functional Requirements (FR)

- **FR1:** The system must allow clients and staff to authenticate securely using email and password.
- **FR2:** The client must be able to view available funeral service packages, casket specifications, and add-on pricing on Flutter.
- **FR3:** The client must be able to submit a funeral booking request with deceased details and preferred burial dates on Flutter.
- **FR4:** The client must be able to view real-time arrangement status notifications on their mobile application.
- **FR5:** The parlor administrator must be able to review, approve, reject, or modify booking requests via ASP.NET Core.
- **FR6:** The administrator must be able to assign parlor assets (hearse, driver, tent) to approved bookings without date conflicts.
- **FR7:** The administrator must be able to search policy records and log client monthly premium payments on the web portal.
- **FR8:** The web application must generate daily schedule reports for mortuary and transport staff.

9.2 Non-Functional Requirements (NFR)

- **NFR1 (Security):** Passwords must be hashed using bcrypt/Argon2 via Supabase Auth; all API requests require valid JWT tokens.
- **NFR2 (POPIA Privacy):** System must strictly isolate records via PostgreSQL RLS; synthetic test data must be used for prototype runs.
- **NFR3 (System Latency):** Booking submissions in Flutter must reflect on the ASP.NET dashboard within 3.0 seconds under broadband.
- **NFR4 (Usability):** Mobile UI must provide simplified navigation and high contrast typography tailored for grieving users.

9.3 System & Integration Requirements

- **SR1:** Mobile client developed in Flutter SDK (Dart); Web portal in ASP.NET Core MVC (C# / .NET 8).
- **SR2:** Shared backend hosted on Supabase (PostgreSQL 15+); code versioned on GitHub.
- **IR1:** Bookings submitted in Flutter must sync instantly to Supabase and stream to the ASP.NET dashboard.
- **IR2:** Status updates made in ASP.NET must propagate to the Flutter mobile client via Supabase Realtime

10. REQUIREMENTS PRIORITIZATION & TRACEABILITY MATRIX

ID	Type	Requirement Description	Priority	Related Stakeholder Need
FR1	FUNCTIONAL	User registration & role-based authentication	MUST HAVE	Access Control & Security
FR2	FUNCTIONAL	Mobile funeral package & casket catalog viewer	MUST HAVE	Package & Price Transparency
FR3	FUNCTIONAL	Mobile funeral service booking request intake	MUST HAVE	Remote Service Booking
FR4	FUNCTIONAL	Mobile real-time booking status tracker card	MUST HAVE	Booking Calendar Control
FR5	FUNCTIONAL	Admin booking approval/rejection console	MUST HAVE	Booking Calendar Control
FR6	FUNCTIONAL	Hearse, driver & equipment dispatch allocator	MUST HAVE	Conflict-Free Scheduling
FR7	FUNCTIONAL	Burial society policy lookup & premium payment logger	MUST HAVE	Policy Record Management
FR8	FUNCTIONAL	Daily dispatch schedule reports for staff	SHOULD HAVE	Operational Coordination
NFR1	NON-FUNCTIONAL	Password encryption & JWT session tokens	MUST HAVE	Data Protection Standard
NFR2	NON-FUNCTIONAL	POPIA data isolation via PostgreSQL RLS	MUST HAVE	Customer Privacy Compliance
SR1	SYSTEM	Flutter mobile client and ASP.NET Core web portal	MUST HAVE	Curriculum tech stack
SR3	SYSTEM	Supabase shared database & authentication	MUST HAVE	Curriculum tech stack
IR1	INTEGRATION	Instant booking sync from Flutter to Supabase to ASP.NET	MUST HAVE	Automated Booking Intake
IR2	INTEGRATION	Status updates propagation from ASP.NET to Flutter	MUST HAVE	Real-time Client Notifications

11. FEASIBILITY ANALYSIS (SIX-DIMENSION EVALUATION)

Feasibility Area	Evaluation & Justification	Evaluation Status
Technical Feasibility	The development team has acquired C# (.NET) and Dart (Flutter) programming competencies through diploma coursework. Official Supabase SDKs provide direct integration libraries for Flutter and ASP.NET Core. Student development laptops support the full development environment.	**FEASIBLE**
Operational Feasibility	The prototype directly solves Mangojane's core operational pain points by eliminating paper booking diary clashes and enabling remote service viewing. User interfaces are designed for straightforward adoption by non-technical clients and parlor administrators.	**FEASIBLE**
Economic Feasibility	The project utilizes open-source development tools and free cloud service tiers (Supabase Free Tier, GitHub Student Plan). Zero software licensing or infrastructure costs will be incurred by the university, team, or parlor.	**FEASIBLE**
Schedule Feasibility	The 14-week semester timeline allows structured execution across Phase 1 (Weeks 2–6), Phase 2 (Weeks 7–9), and Phase 3 (Weeks 10–14). The baseline scope is focused on essential core features to guarantee completion.	**FEASIBLE**
Legal Feasibility	Fully compliant with the Protection of Personal Information Act (POPIA). Fictionalized and synthetic test records will be utilized across development, testing, and demonstration, preventing exposure of actual citizen data.	**FEASIBLE**
Ethical Feasibility	Bereavement services manage sensitive human grief. The system protects family dignity by enforcing strict access control, securing client communications, and providing clear, transparent pricing structures.	**FEASIBLE**

Overall Feasibility Verdict: FEASIBLE. The technical architecture, scope boundaries, and development schedule are realistic and achievable within the semester constraints.

12. PROJECT RISK REGISTER (TOP 10 RISKS)

ID	Risk Statement	Likelihood	Impact	Score	Mitigation Strategy	Risk Owner	Status
RSK-01	Exposure of Supabase API service role keys on public GitHub repo.	**Low**	**Critical**	**High**	Enforce .gitignore and local .env variables; conduct automated credential scans.	QA Specialist	Active
RSK-02	Schema mismatches between ASP.NET models and Flutter DTOs.	**High**	**Medium**	**High**	Design Supabase tables centrally first; conduct weekly schema syncs.	Backend Lead	Active
RSK-03	Scope creep through complex payment gateway integration requests.	**Medium**	**High**	**High**	Formally document live payment as out-of-scope; use mock payment reference IDs.	Lead Developer	Mitigated
RSK-04	Team members lack Flutter state management proficiency.	**Medium**	**High**	**High**	Pair junior developers with Flutter lead; utilize standard Provider/ setState architecture.	Mobile Lead	Active
RSK-05	Stakeholder unavailability during mid-semester design reviews.	**Medium**	**Medium**	**Medium**	Establish structured asynchronous progress notes and pre-booked review sessions.	Requirements Lead	Active
RSK-06	Database connection pool exhaustion on Supabase free-tier backend.	**Low**	**Medium**	**Low**	Properly dispose of database connections in ASP.NET and Flutter clients.	Backend Lead	Active
RSK-07	Git merge conflicts across 7 developers working on overlapping code.	**High**	**Medium**	**Medium**	Enforce branch-per-feature workflow; require pull request code reviews.	QA Specialist	Active
RSK-08	Incomplete task progress logs resulting in contribution disputes.	**Medium**	**High**	**High**	Enforce weekly updates to Project Progress Tracker before group standups.	Lead Developer	Active
RSK-09	Hardware or network connectivity failure during final demonstration.	**Medium**	**High**	**High**	Prepare fallback demonstration video and run local Supabase emulator fallback.	Web Lead	Active
RSK-10	Accidental inclusion of real stakeholder customer contact data.	**Low**	**High**	**High**	Sanitize all datasets; utilize exclusively synthetic mock customer profiles.	Requirements Lead	Mitigated

13. REQUIREMENTS CHANGE LOG & SCOPE REFINEMENT

Change ID	Proposed Modification	Reason / Justification	Impact on Scope	Decision
CR-01	Omit live banking payment gateway integration.	High technical complexity, merchant account requirement, and security risk for prototype.	Scope reduced; simulated via mock payment receipts.	Approved
CR-02	Omit live DHA Home Affairs sync for death notices.	Requires government API clearances unavailable to academic prototypes.	Replaced with manual document attachment upload.	Approved
CR-03	Omit hardware-level GPS vehicle tracking.	Hardware costs, equipment installation constraints, and limited semester time.	Replaced with manual administrative status dispatch selector.	Approved

14. APPENDICES: STAKEHOLDER EVIDENCE & SETUP

Appendix A: Stakeholder Interview Questions and Answer

1. How is a funeral arrangement handled from first contact to burial?
2. What problems are experienced with the current paper diaries?
3. Which issue causes the greatest operational bottleneck?
4. What features should bereaved families see on the mobile app?
5. What administrative tools do staff need on the web portal?
6. How are burial society policies and premium payments verified?
7. What daily schedule reports do drivers and mortuary staff need?
8. What data privacy considerations apply to customer records?

Appendix B: Recorded Stakeholder Responses & Interview Findings

Question Area	Stakeholder Response Summary (Bongani Hospital OPD & IT)	Derived Prototype Requirement
Current Booking Process	The family calls our landline or comes in person. We write all details in a paper diary. We confirm the date, then call burial society to verify membership. On the day, driver and mortuary staff check the diary. After burial we file the papers in a box.	Remote client intake and digital booking registration (FR1, FR3) to replace paper diaries.
Diary Bottlenecks & Clashes	Big problems . We lose diary sometimes. We have double bookings because too staff write at once. We can not search old clients. If a family calls at night we cannot check prices.	Centralized conflict-free booking queue and asset allocation calendar (FR5, FR6).
Operational Disruption	"Phone calls. We get 20 to 30 calls everyday asking 'How much is a coffin?' 'Are you free on Saturday?'. One person must sit and answer phones instead of helping families."	Transparent mobile package & casket catalog viewer (FR2) with pricing visibility.
Mobile Client Needs	"They must see our service packages and prices. They must book a funeral online. They must check if the burial society is active. They must see address, directions and contact us on WhatsApp. No need to call at 2am."	Self-service booking submission, live milestone tracker card, and society check (FR2, FR3, FR4).
Web Management Tools	"We need a dashboard to see all bookings. Calendar for drivers and mortuary. We need to track payments and burial society premiums. We need to print invoices and daily reports. We need to add / edit accounts.".	Administrative dashboard, resource dispatch calendar, and premium payment logger (FR5, FR6, FR7).
Policy Verification	"Right now we phone the burial society office and wait 1 to 2 days for reply. Sometimes they don't answer. We need the system to check status online automatically."	Centralized policy lookup and member payment status verification engine (FR7).
Daily Operational Reports	"Every morning they need a list. Name of deceased, pickup time, pickup address, church and cemetery. They also need vehicle assigned and contact person. Printed or on their phones."	Automated daily dispatch schedule report generation for transport and mortuary crews (FR8).
POPIA & Data Privacy	"Customer ID numbers, cause of death, and family contact details are private. Only admin and owner must see full details. Staff must login with password. We must comply with POPIA because it is sensitive information."	Role-Based Access Control (RBAC) and Supabase Row-Level Security (RLS) data isolation (NFR1, NFR2).

Appendix C: Evidence of Stakeholder Engagement & Follow-Up Log

Engagement Record Summary:

- **Stakeholder Facility:** Mangojane Funeral Parlor, Welkom, Free State
- **Initial Engagement:** 28 August 2026 | 12:00 – 13:30 SAST | Parlor Operations Office.
- **Participants:** Hlompho Compact Malejane (Lead Facilitator), Nomsa Nomankonya (Note-Taker), Parlor General Manager & Operations Supervisor
- **Requirements Confirmation:** 26 August 2026 | In-person follow-up session reviewing requirements matrix and charter.
- **Current Status:** Operational workflow and system requirements verified; physical confirmation sheet signed on-site.

Appendix D: Stakeholder Confirmation & Sign-Off Status

Role / Entity	Name & Designation	Status / Verification
External Stakeholder	Lerato Moipone Mongojane(Co-Founder and Manager)	Requirements Confirmed & Signed (28/08/2026)
Project Lead / Facilitator	Hlompho Compact Malejane (The Extended Group)	Baseline Compiled (28/08/2026)
ITC327W Supervisor	MR. Mokoduwe / Faculty of ICT, CUT	Unit 1 & 2 Formative Checkpoint

Appendix E: Initial Supabase & GitHub Project Setup Evidence

Supabase Cloud Backend Project Confirmation

Project Name: Mangojane-Funeral-management-prototype | Region: af-south-1 | PostgreSQL Version: 15.x

[SUPABASE CLOUD PROJECT DASHBOARD]
Organization : CUT-ITC327W-TheExtendedGroup
Project Name : Mangojane-Funeral-management-prototype
Status : ACTIVE (Healthy)
Database : PostgreSQL (RLS Enabled)
Realtime : public.bookings & public.policy_payments
API Endpoint(superbase) : https://gfxooglajripoikobnec.supabase.co
GitHub Repo : https://github.com/CHmale/Mangojane-Funeral-Management-System

CHmale

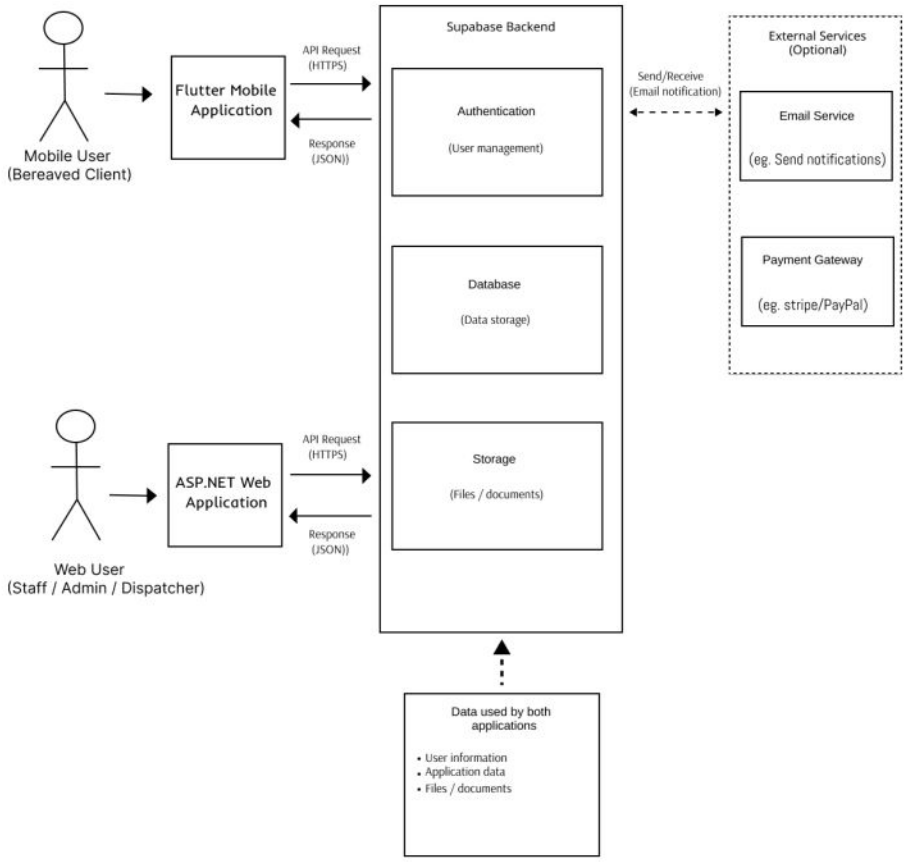
Lead Developer / Facilitator

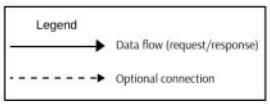
Stakeholder (Mangojane Parlor)

PART II:PHASE 2 SYSTEM ARCHITECTURE AND DETAILED DESIGN

15. System Architecture Design

The Mangojane Integrated Funeral Management System is organized as a robust, decoupled three-tier architecture connecting the presentation clients with a managed backend layer and relational data storage. The entire infrastructure is hosted within the AWS Cape Town cloud zone (af-south-1) to guarantee low network latency and enforce South African POPIA data residency compliance.





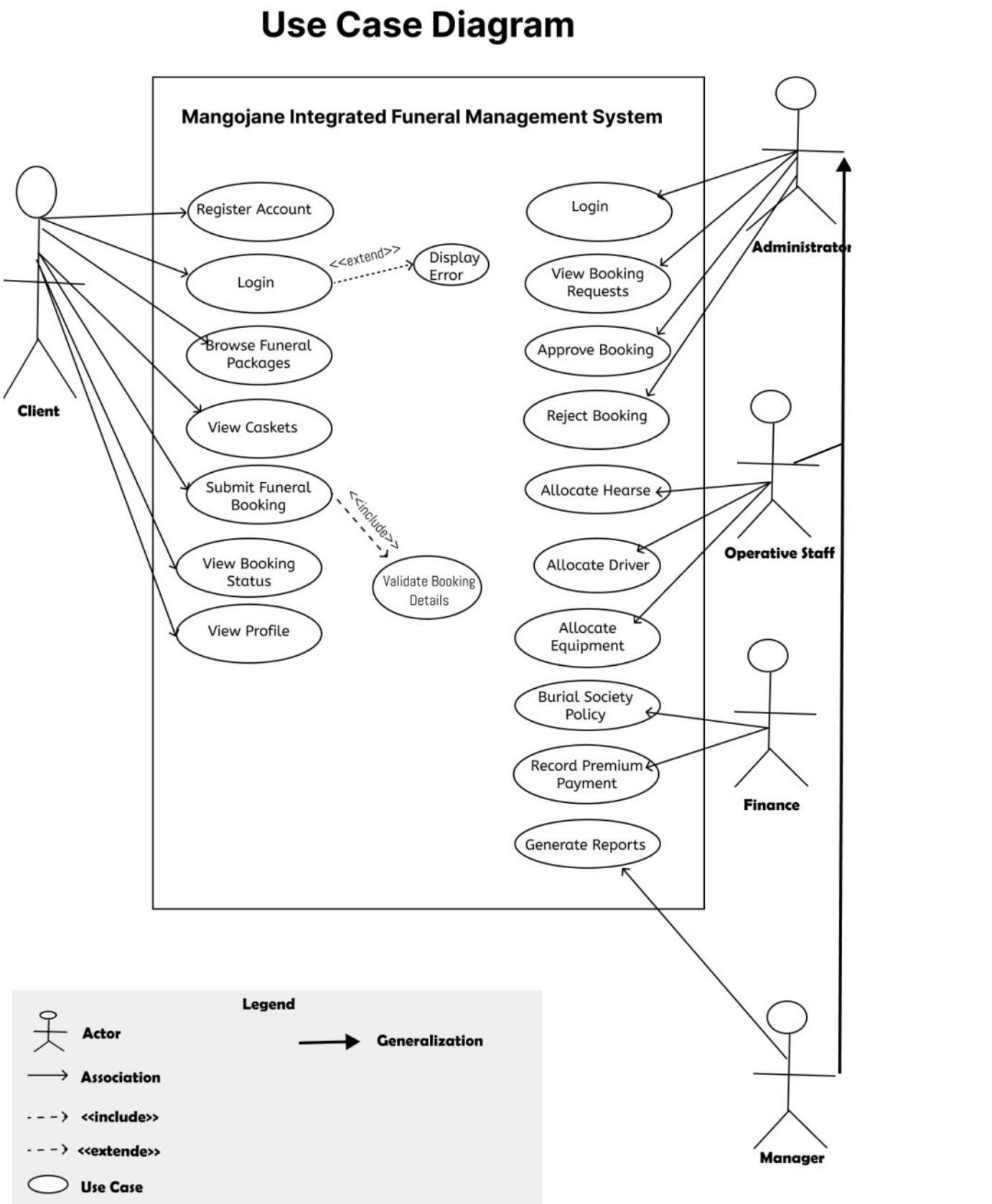
Architectural Subsystem Responsibilities

- Flutter Mobile Client (Dart): Serves bereaved family members and policyholders. Provides a responsive mobile interface optimized for low bandwidth, allowing clients to register, explore official service tiers, submit booking requests, upload supporting documents, and receive live arrangement status notifications.
- ASP.NET Core 8 MVC Web Portal (C#): Serves parlor administrators, dispatch coordinators, and parlor ownership. Features a centralized booking triage queue, a conflict-free asset allocation calendar, a burial society policy and premium transaction logger, and automated PDF daily route dispatch sheet generation.
- Supabase Cloud Backend (PostgreSQL 15, GoTrue, Realtime): Unifies authentication via signed JSON Web Tokens (JWT), provides real-time change data capture through WebSockets (WSS), and enforces multi-tenant Row-Level Security (RLS) directly inside the database engine.

16. UML Behavioral Modeling

16.1 Use Case Modeling & Actor Generalization

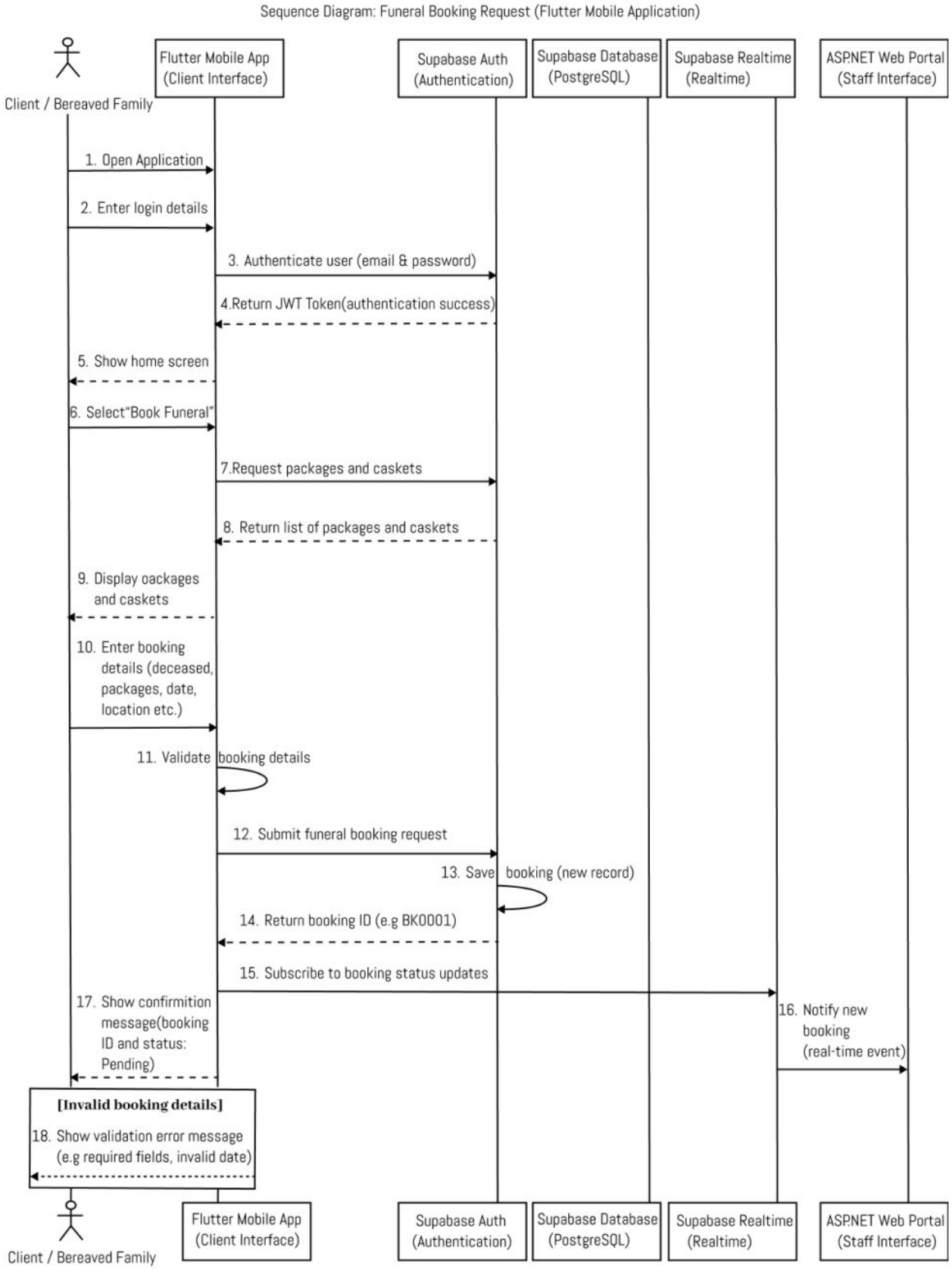
The behavioral specification comprises five actors interacting with 10 structured use cases within the Mangojane System Boundary. Mangojane staff actors (Parlor Administrator, Fleet Dispatcher, and Parlor Owner) specialize the base Mangojane Staff actor through a standard UML generalization hierarchy, inheriting fundamental system authentication and operational visibility.



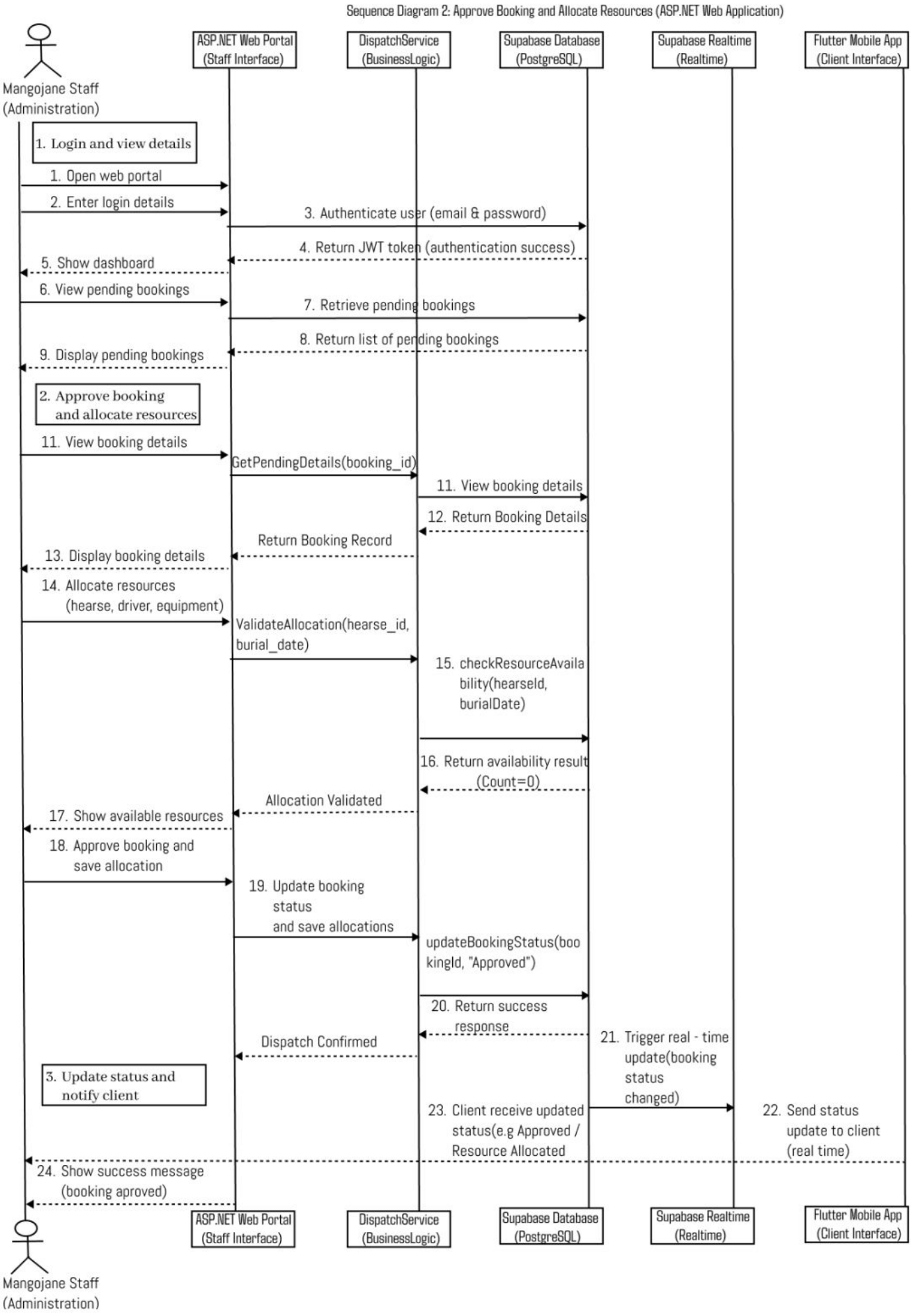
16.2 Dynamic Process Sequence Diagrams

Two comprehensive dynamic sequence diagrams model the mission-critical workflows of the system:

- **Sequence Diagram 1 (Mobile Funeral Booking Intake):** Illustrates the end-to-end flow from client authentication on the Flutter app, package retrieval from Supabase, local client validation, record insertion into FUNERAL_BOOKING, and real-time event broadcasting to the ASP.NET administrative queue.



- Sequence Diagram 2 (Administrative Approval & Asset Dispatch): Details parlor administrator login on ASP.NET, pending queue inspection, hearse and driver allocation, backend collision verification against uq_hearse_booking_date, database commit, and instant WebSocket push notification back to the mobile client milestone card.



17. Database Design: Entity-Relationship Diagram (ERD) & Schema

The system database was designed and verified using Oracle SQL Developer Data Modeler and fully normalized to Third Normal Form (3NF). The schema contains six core relational entities structured to prevent redundant storage and eliminate booking collisions.

Entity Name	Primary Roles & Responsibilities	Interface Used	Core Needs & Constraints
CUSTOMER	CustomerID (INT)	None	FullName, Email (Unique), PhoneNumber, Role (Client/ AdmiCreatedAt. Enforces client identity.
SERVICE_PACKAGE	PackageID (INT)	None	PackageName (Bronce, Silver, Gold,Platinum),CasketModel,Base Price, Inclusions (CLOB). Official rate card catalog
VEHICLE_ASSET	VehicleID (INT)	None	RegistrationNumber (Unique), VehicleType (Hearse, FamLimousine, Truck), OperationalStatus.
FUNERAL_BOOKING	BookingID (INT)	CUSTOMER_CustomerID, SERVICE_PACKAGE_PackageID,VEHICLE_ASSET_VehicleID	DeceasedName, DeceasedIDNumber (13-digit), BurialDatePickupAddress, CemeteryName, Status. Unique constraint uq_hearse_booking_date(VEHICLE_ASSET_VehicleID,BurialDate) prevents double-bookings.
POLICY	PolicyID (INT)	CUSTOMER_CustomerID	PolicyNumber (Unique), PlanType, MonthlyPremium, Stat(Active, Lapsed), WaitingPeriodExpiry (6-month rule).
POLICY_PAYMENT	PaymentID (INT)	POLICY_PolicyID	PaymentReference (Unique), AmountPaid, PaymentDate, ReceivedBy. Transaction audit trail.

18. User Interface Design & Stakeholder Branding Alignment

Following design review sessions with Ms. Lerato Moipone Mongojane, the system user interfaces were overhauled to embody the official brand colors, dignity, and real-world services of Mangojane Funeral

18.1 Stakeholder Color Palette Integration

Obsidian Black #181513 Primary Background	Espresso Bronze #2B1F17 Card Containers	Mangojane Gold #B7A87C Accents & CTAs	Alabaster White #F7F7F8 Body Typography

18.2 Authentic Service Packages & Rate Cards

- **Bronce Package (R180 / month):** Single member cover, standard redwood flat-lid casket, primary hearse, and 1 family escort vehicle.
- **Silver Package (R230 / month):** Family cover, 3-tier pine casket, primary hearse, 2 family cars, and family gravesite tent.
- **Gold Package (R280 / month):** Extended family cover, kiaat flat-lid casket, luxury hearse, 3 family cars, tent, chairs, and 100 printed funeral programs.
- **Platinum Package (R350 / month):** Executive comprehensive cover, executive quarterview redwood casket, luxury hearse, executive family fleet, sound equipment, and complete cemetery setup.

18.3 Designed Interface Suites

- **Flutter Mobile Client (17 Screens designed in Figma):** Splash Screen, Onboarding, Login, Registration, Dashboard, Package Catalog, Package Details (with real rate cards), 3-Step Booking Intake Wizard, Document Attachment Uploader, Booking Confirmation, Live Arrangement Milestone Tracker Card, Policy Information View, Payment Receipt History, Branch Directory (Thabong and Wesselsbron), and Profile / POPIA Privacy Settings**.**
- **ASP.NET Core Web Portal (4 Core Views designed in Figma):** Administrative Staff Login, Real-Time Booking Triage Queue, Fleet & Mortuary Scheduler Calendar (with automated collision alerts), and Policy Ledger / Daily Route Sheet PDF Export Engine.

19. Design Alignment & Traceability Matrix (5-Way Verification)

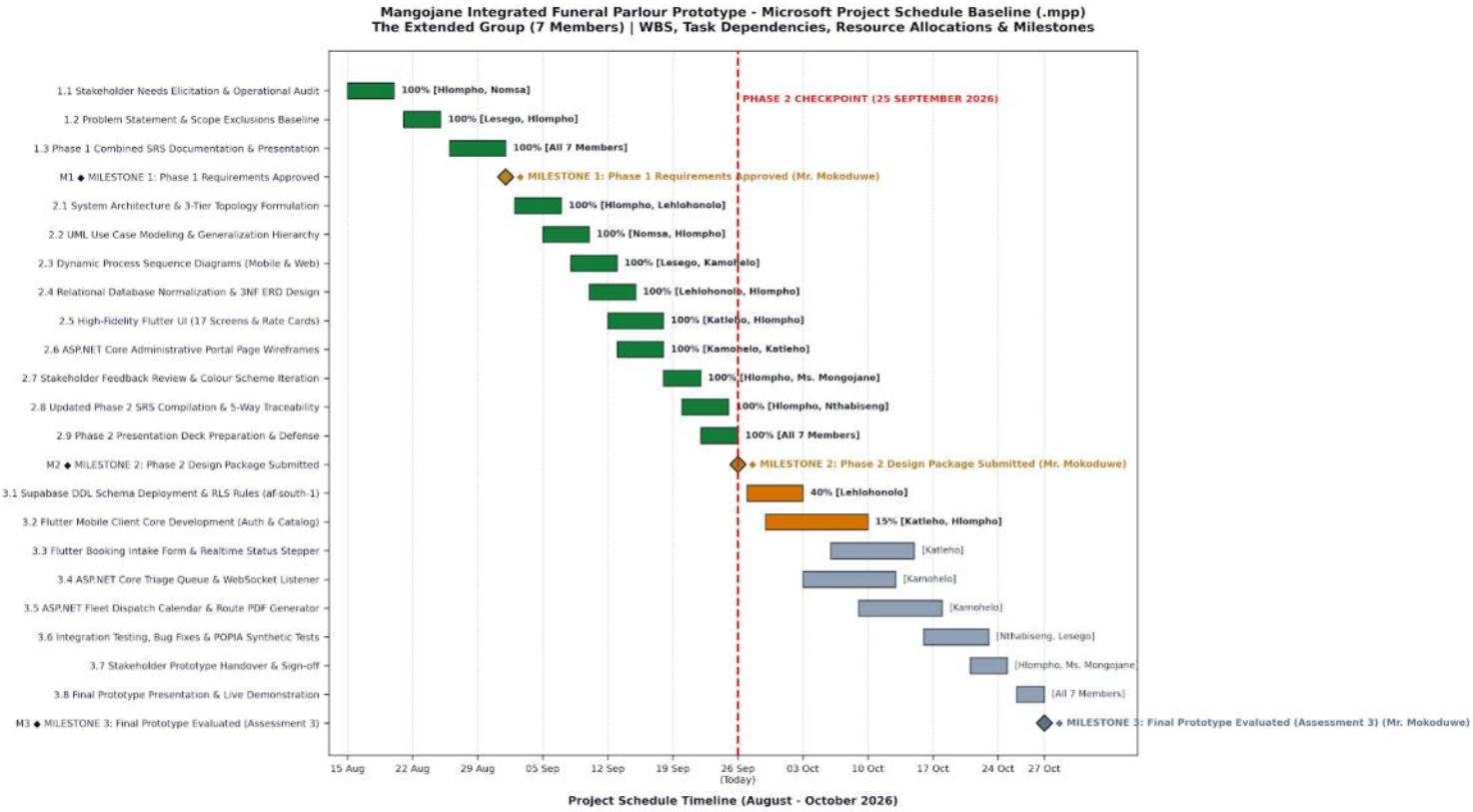
Req ID	Requirement Description	UML Use Case	Sequence Diagram	ERD Table & Constraints	Primary UI Screens
FR1	User authentication via email/password	**UC01, UC01a**	**Seq 1 (2–4), Seq 2 (2–4)**	CUSTOMER (Email, Role)	SCR-MOB-03, SCR-WEB-01
FR2	Browse packages, caskets, and pricing	**UC02**	**Seq 1 (7–9)**	SERVICE_PACKAGE	SCR-MOB-06, SCR-MOB-07
FR3	Submit funeral booking with deceased info	**UC03, UC04**	**Seq 1 (10–14)**	FUNERAL_BOOKING	SCR-MOB-08, 09, 10, 11
FR4	Real-time booking milestone notifications	**UC05**	**Seq 1 (15), Seq 2 (21–23)**	FUNERAL_BOOKING (Status), WSS	SCR-MOB-13 (Tracker Card)
FR5	Admin review, triage, approve/reject queue	**UC06**	**Seq 2 (6–13, 18–20)**	FUNERAL_BOOKING	SCR-WEB-02 (Triage Queue)
FR6	Conflict-free hearse & asset dispatch	**UC07**	**Seq 2 (14–17)**	VEHICLE_ASSET, uq_hearse_booking_date	SCR-WEB-03 (Scheduler)
FR7	Policy search & monthly premium logger	**UC08**	**Admin Direct POST**	POLICY, POLICY_PAYMENT	SCR-MOB-14, SCR-WEB-04
FR8	Daily staff dispatch route sheets	**UC09**	**Reporting Query**	BOOKING $ owtie$ VEHICLE $ owtie$ PACKAG	SCR-WEB-04 (Route PDF)

20. Original Design Source Links & Tool Specifications

Design Deliverable	Design Tool Used	Tool Type	Source Link / Editable File Name	Responsible Author
EntityRelationship Diagram (ERD	Oracle SQL Developer Data Modeler	Desktop SQL IDE	N/A	Hlompho Malejane
System Architecture Diagram CR-03	Figma Design	Cloud Collaborative	https://www.figma.com/design/LwliOD2Hn8I4YxUOHKpqUK/Untitled?node-id=0-1&t=miJwqytSUbCq9hnm-1	Lehlohonolo Modise
UML Use Case Diagram	Figma Design	Cloud Collaborative	https://www.figma.com/design/TSOIluVPJlyfYcI8Z25R9o/Untitled?node-id=19-222&t=miEFIzS7NR1X3E32-1	Nomsa Nomankonya
Dynamic Sequence Diagrams (1 & 2)	Figma Design	Cloud Collaborative	https://www.figma.com/design/LwliOD2Hn8I4YxUOHKpqUK/Untitled?node-id=0-1&t=miJwqytSUbCq9hnm-1	Lesego Waeza
17 Flutter Mobile Screens	Figma Design	Cloud Collaborative	https://www.figma.com/design/9iZas2zsr7EDfPw2DOU6Tv/221037680?t=jAa8RNzSOr7uMzW0-1	Katleho Mokhele
ASP.NET Core Web Views	Figma Design	Cloud Collaborative	https://www.figma.com/design/ayXmUcQVYWhE7QX1e7vHnL/Mangojane-Funeral-Services---ASP.NET-Staff-Admin-Prototype-FINAL?t=jAa8RNzSOr7uMzW0-1	Kamohelo Mohlohlo
Microsoft Project Schedule	Microsoft Project	Online Platform	https://planner.cloud.microsoft/ui/web/v1/plan/jB-Ml-2m_Uep84RDD7WB-WQAAVyy?tid=b940c149-91b7-4bcf-ba2c-0db8d817196d	Nthabiseng Moeketsi

21. Microsoft Project Schedule & Phase 2 Baseline Status

The baseline schedule in Microsoft Project (.mpp) tracks 100% completion across all Unit 1 requirements and Unit 3 system design deliverables, with implementation activities currently underway for Phase 3.



22. Phase 2 Stakeholder Re-Validation & Engagement Log

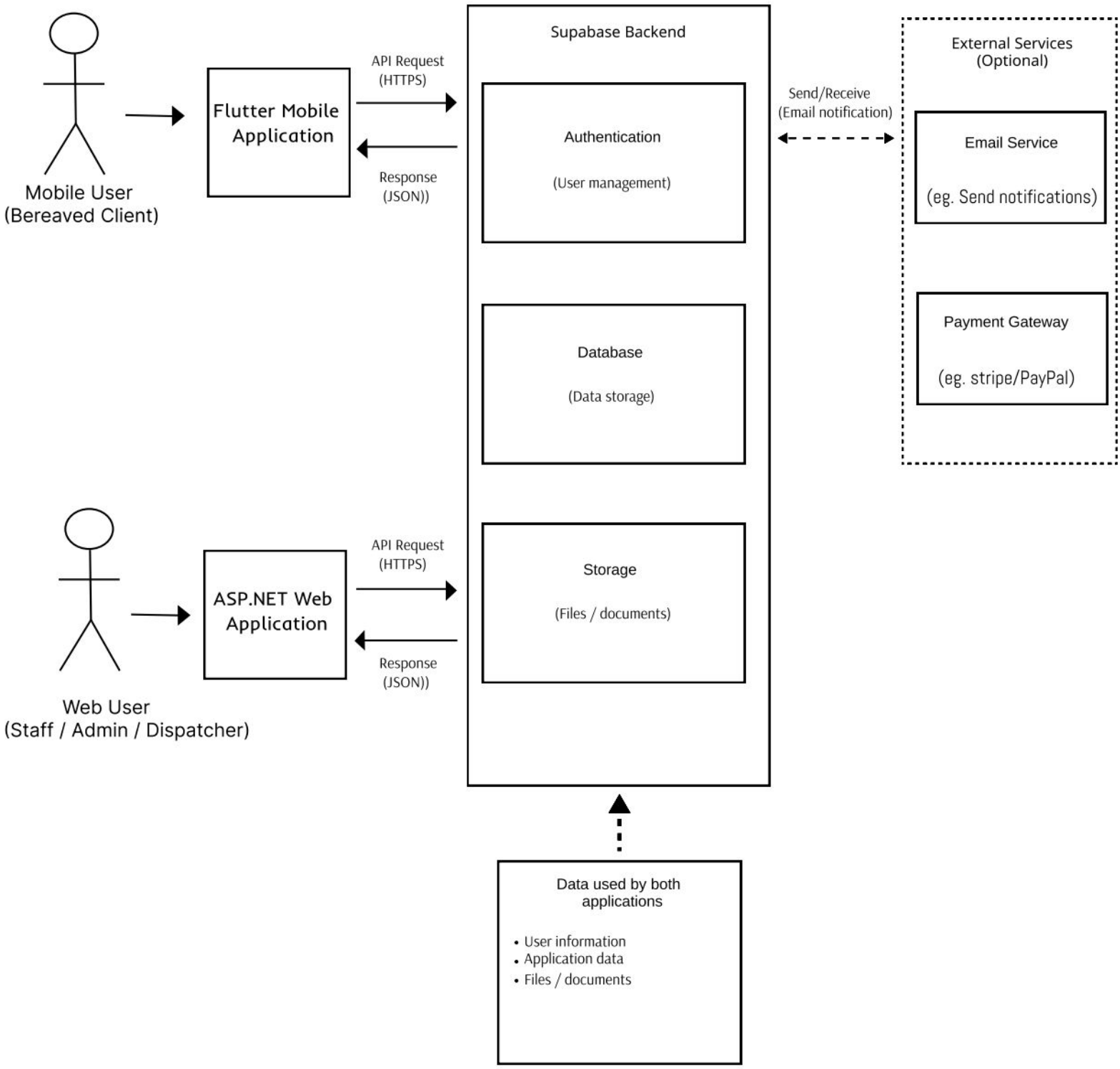
- **Log ID SE-03 (14 September 2026):** Meeting at Mangojane Parlor Office (Thabong) with Ms. Lerato Moipone Mongojane. Reviewed initial wireframes. The stakeholder instructed the team to replace generic blue screens with their authentic corporate colors (Obsidian Black, Warm Bronze, Gold) and ensure service offerings match their official rate cards. All 17 Flutter screens and 4 ASP.NET views were updated accordingly in Figma.
- **Log ID SE-04 (22 September 2026):** Field review of Oracle SQL ERD models and revised Flutter UI mockups. Stakeholder validated all 6 database entities and confirmed operational alignment

23. Phase 2 Engineering Reflection

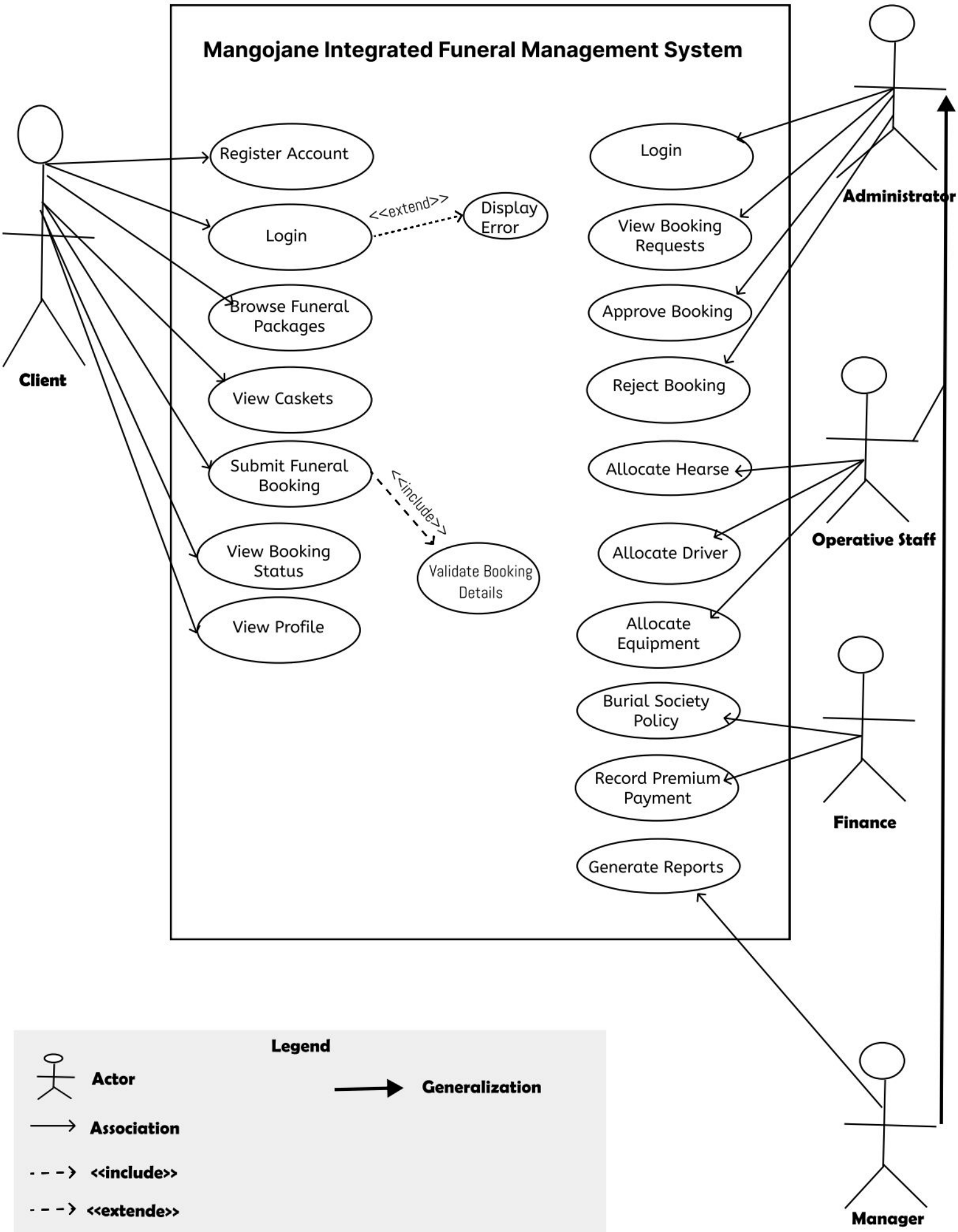
Phase 2 successfully transformed qualitative stakeholder requirements into concrete, verified engineering artifacts. Developing the relational ERD in Oracle SQL Developer Data Modeler guaranteed 3NF normalization, preventing operational double-bookings through database constraints. Designing all system diagrams and interfaces collaboratively in Figma established clear visual guidelines for the upcoming Phase 3 coding sprints.

APPENDIX: SYSTEM DESIGN ARTIFACTS & DIAGRAMS

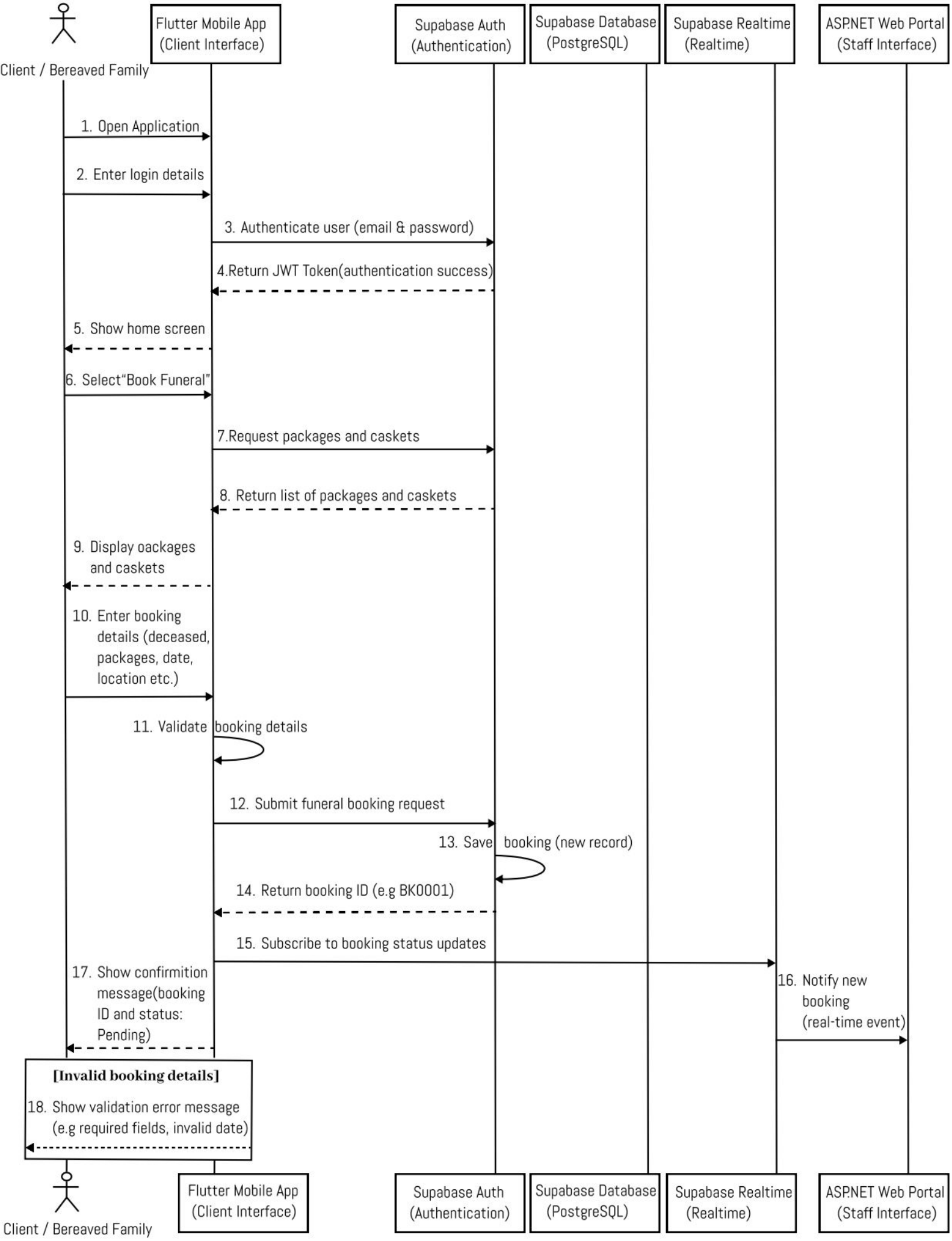
System Architect



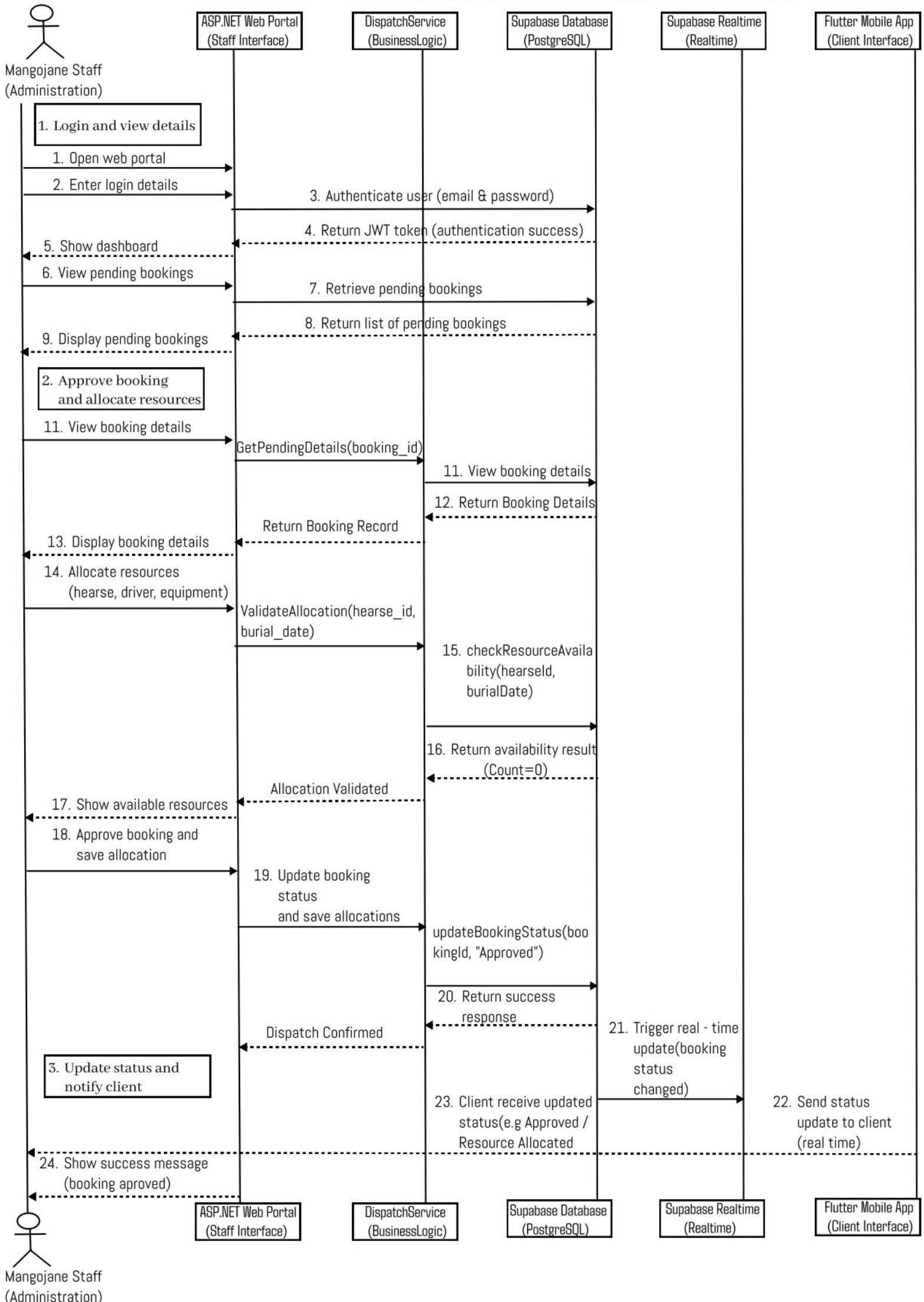
Use Case Diagram

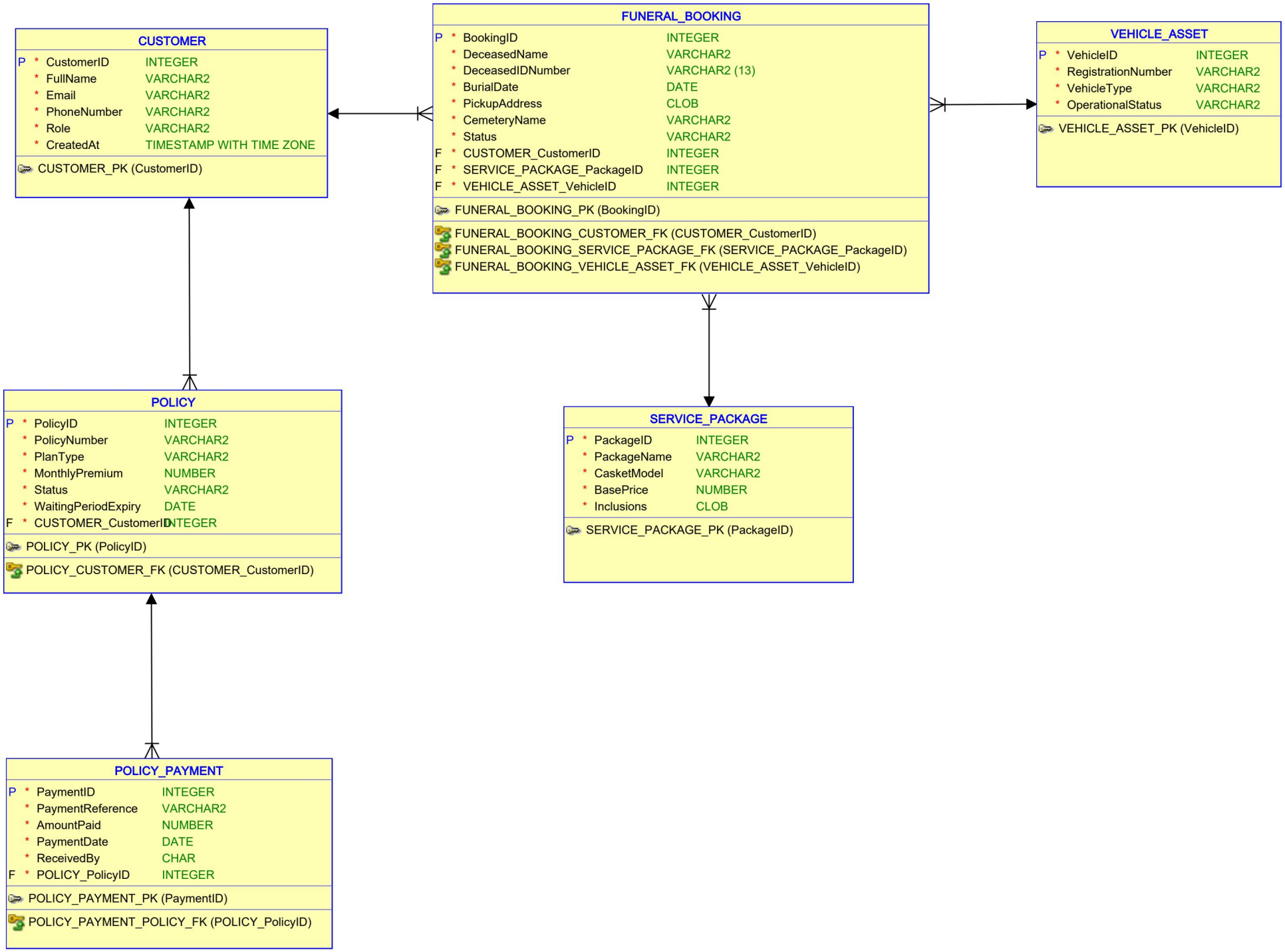


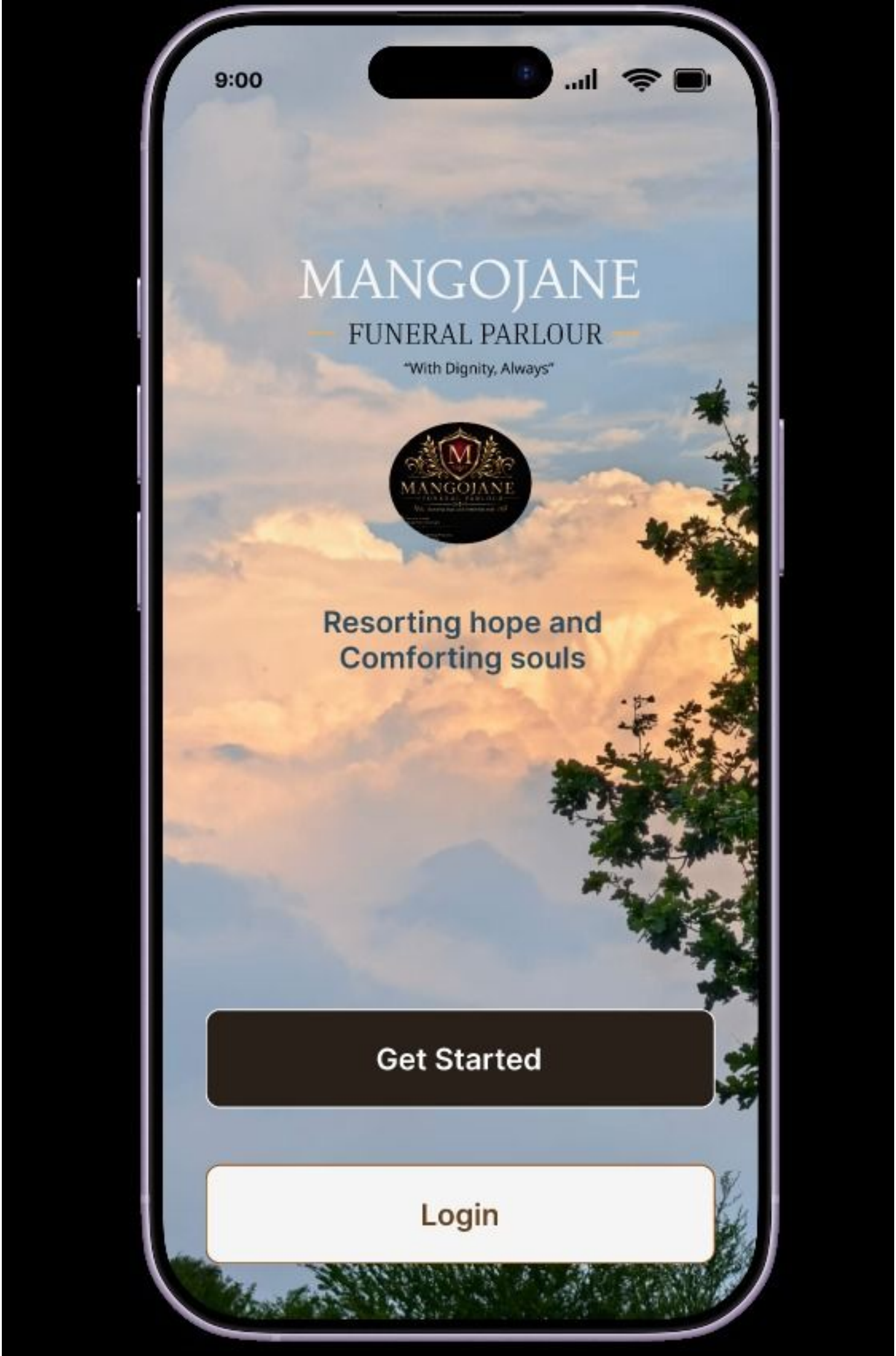
Sequence Diagram: Funeral Booking Request (Flutter Mobile Application)

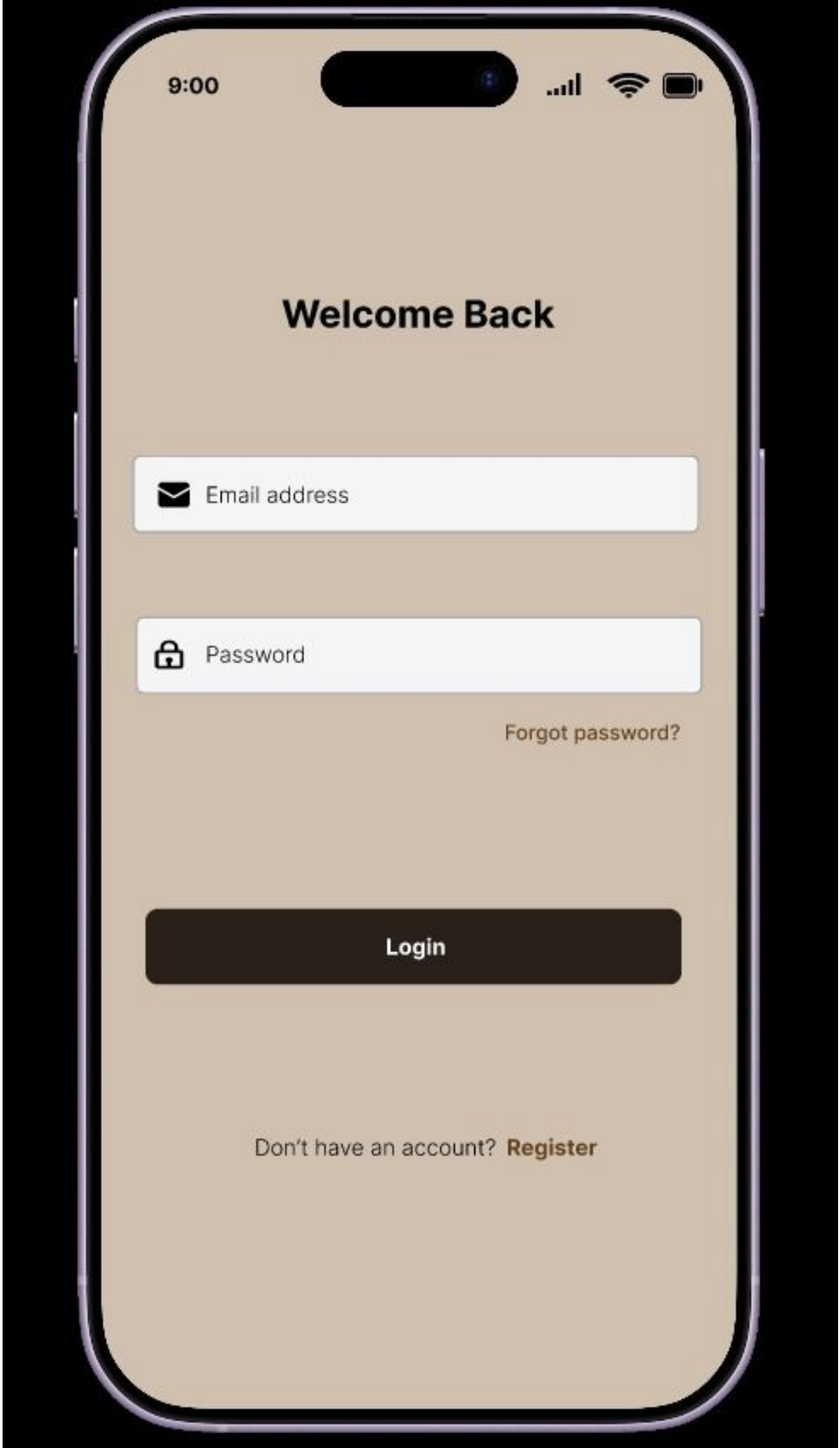


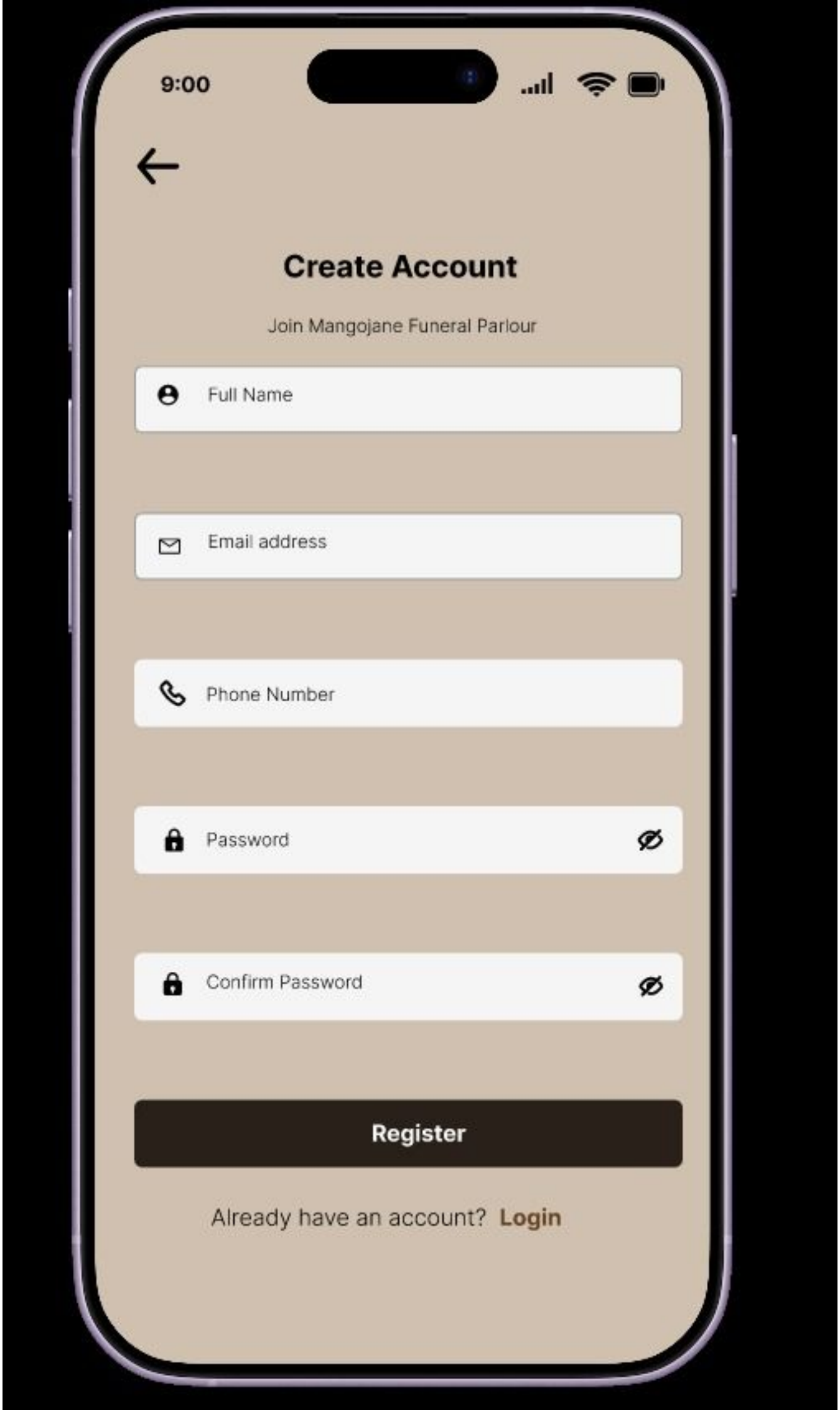
Sequence Diagram 2: Approve Booking and Allocate Resources (ASP.NET Web Application)




CUSTOMER
P * CustomerID INTEGER
* FullName VARCHAR2
* Email VARCHAR2
* PhoneNumber VARCHAR2
* Role VARCHAR2
* CreatedAt TIMESTAMP WITH TIME ZONE
CUSTOMER_PK (CustomerID)
FUNERAL_BOOKING
P * BookingID INTEGER
* DeceasedName VARCHAR2
* DeceasedIDNumber VARCHAR2 (13)
* BurialDate DATE
* PickupAddress CLOB
* CemeteryName VARCHAR2
* Status VARCHAR2
F * CUSTOMER_CustomerID INTEGER
F * SERVICE_PACKAGE_PackageID INTEGER
F * VEHICLE_ASSET_VehicleID INTEGER
FUNERAL_BOOKING_PK (BookingID)
FUNERAL_BOOKING_CUSTOMER_FK (CUSTOMER_CustomerID)
FUNERAL_BOOKING_SERVICE_PACKAGE_FK (SERVICE_PACKAGE_PackageID)
FUNERAL_BOOKING_VEHICLE_ASSET_FK (VEHICLE_ASSET_VehicleID)
VEHICLE_ASSET
P * VehicleID INTEGER
* RegistrationNumber VARCHAR2
* VehicleType VARCHAR2
* OperationalStatus VARCHAR2
VEHICLE_ASSET_PK (VehicleID)
POLICY
P * PolicyID INTEGER
* PolicyNumber VARCHAR2
* PlanType VARCHAR2
* MonthlyPremium NUMBER
* Status VARCHAR2
* WaitingPeriodExpiry DATE
F * CUSTOMER_CustomerID INTEGER
POLICY_PK (PolicyID)
POLICY_CUSTOMER_FK (CUSTOMER_CustomerID)
SERVICE_PACKAGE
P * PackageID INTEGER
* PackageName VARCHAR2
* CasketModel VARCHAR2
* BasePrice NUMBER
* Inclusions CLOB
SERVICE_PACKAGE_PK (PackageID)
POLICY_PAYMENT
P * PaymentID INTEGER
* PaymentReference VARCHAR2
* AmountPaid NUMBER
* PaymentDate DATE
* ReceivedBy CHAR
F * POLICY_PolicyID INTEGER
POLICY_PAYMENT_PK (PaymentID)
POLICY_PAYMENT_POLICY_FK (POLICY_PolicyID)


9:00
MANGOJANE
FUNERAL PARLOUR
"With Dignity, Always"
MANGOJANE
Resorting hope and
Comforting souls
Get Started
Login


9:00
Welcome Back
Email address
Password
Forgot password?
Login
Don’t have an account? Register


9:00
Create Account
Join Mangojane Funeral Parlour
Full Name
Email address
Phone Number
Password
Confirm Password
Register
Already have an account? Login

9:00

Good day,

How can we assist you today?



Browse Packages

View Caskets

Book a Funeral

My Bookings

Home | Packages | Bookings | Profile

9:00

Funeral Packages

All | Standard | Premium | Custom



Bronce Plan

R180.00p/m

Redwood flat lit coffin



Silver Plan

R230.00p/m

Kiaat Flat lit Coffin



Golden Plan

R280.00p/m

3 Tier Kiaat/Pine Coffin



Platinum Plan

R350.00p/m

Quarterview Redwood Casket

9:00

Package Details



Platinum Plan

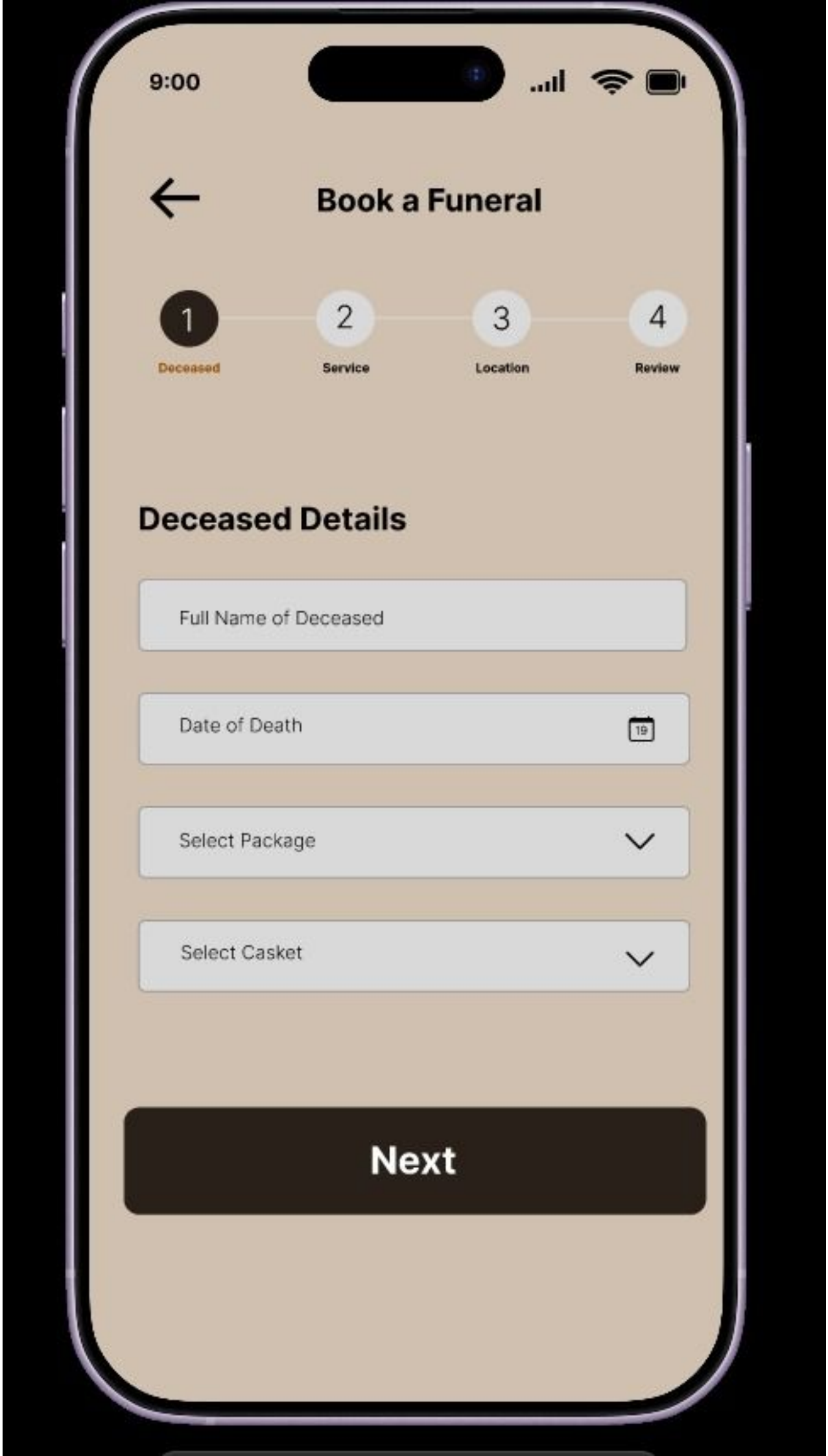
R 350.00p/m

A high-quality Quarterview Redwood Casket with elegant finishes.

Specifications

Dependencies	Up to 5 members
Catering	Optional with added price
Transportation	Family Cars

Select Plan


9:00
Book a Funeral
1
Deceased
2
Service
3
Location
4
Review
Deceased Details
Full Name of Deceased
Date of Death
Select Package
Select Casket
Next

9:00



Booking Submitted

Your funeral booking request has been successfully submitted

Booking Reference

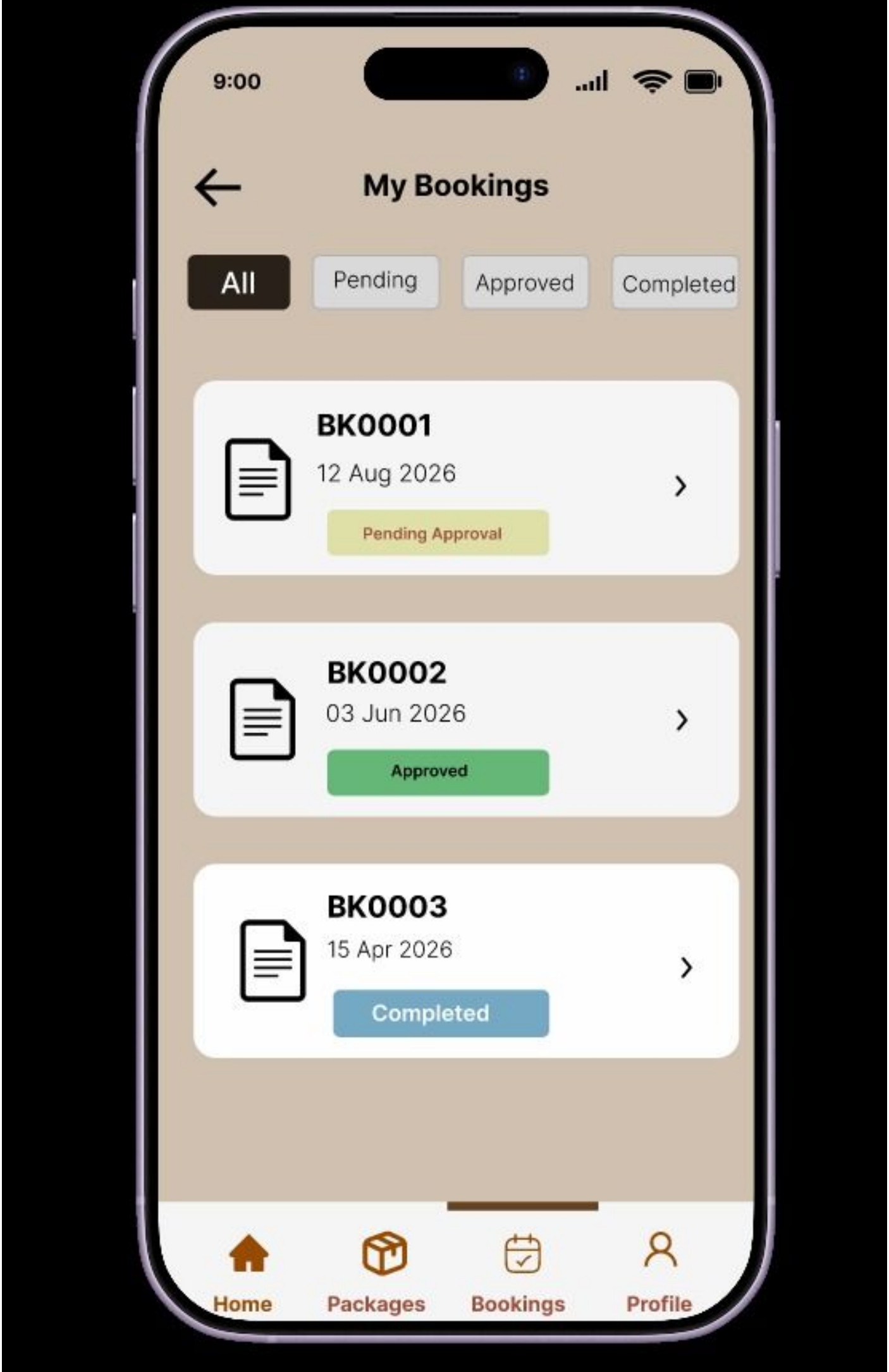
BK0001

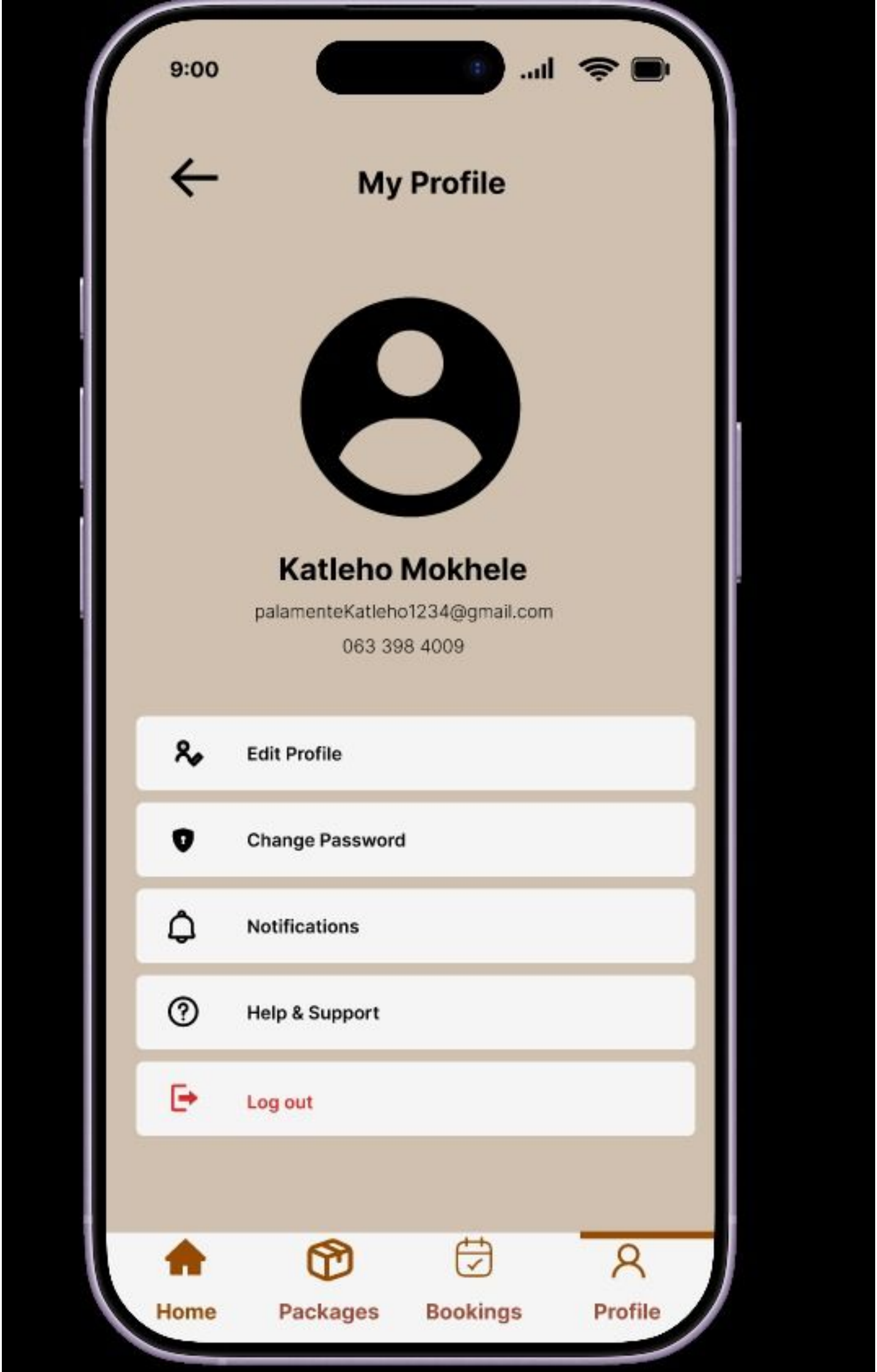
Status **Pending Approval**

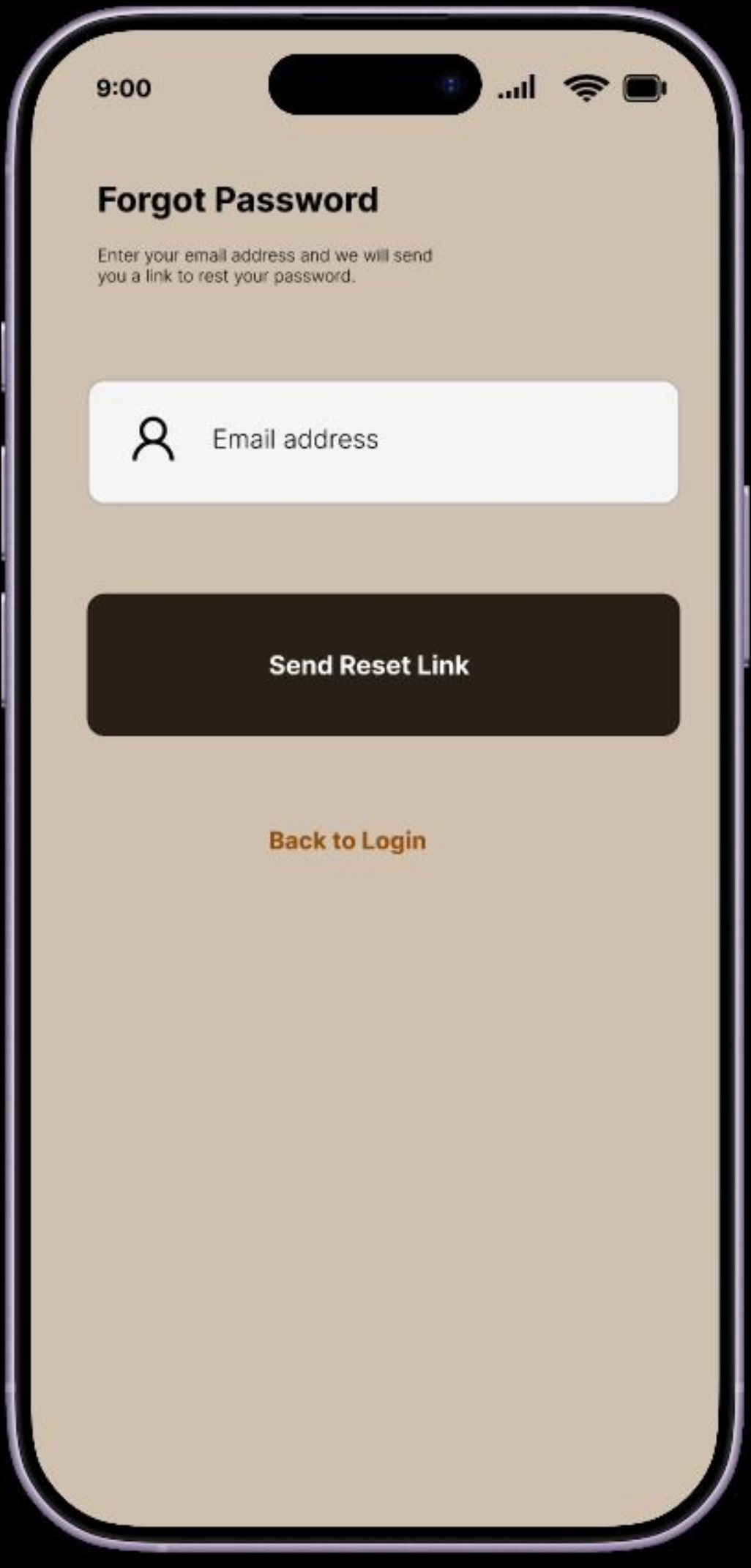
You willl be notified once your booking has been approved.

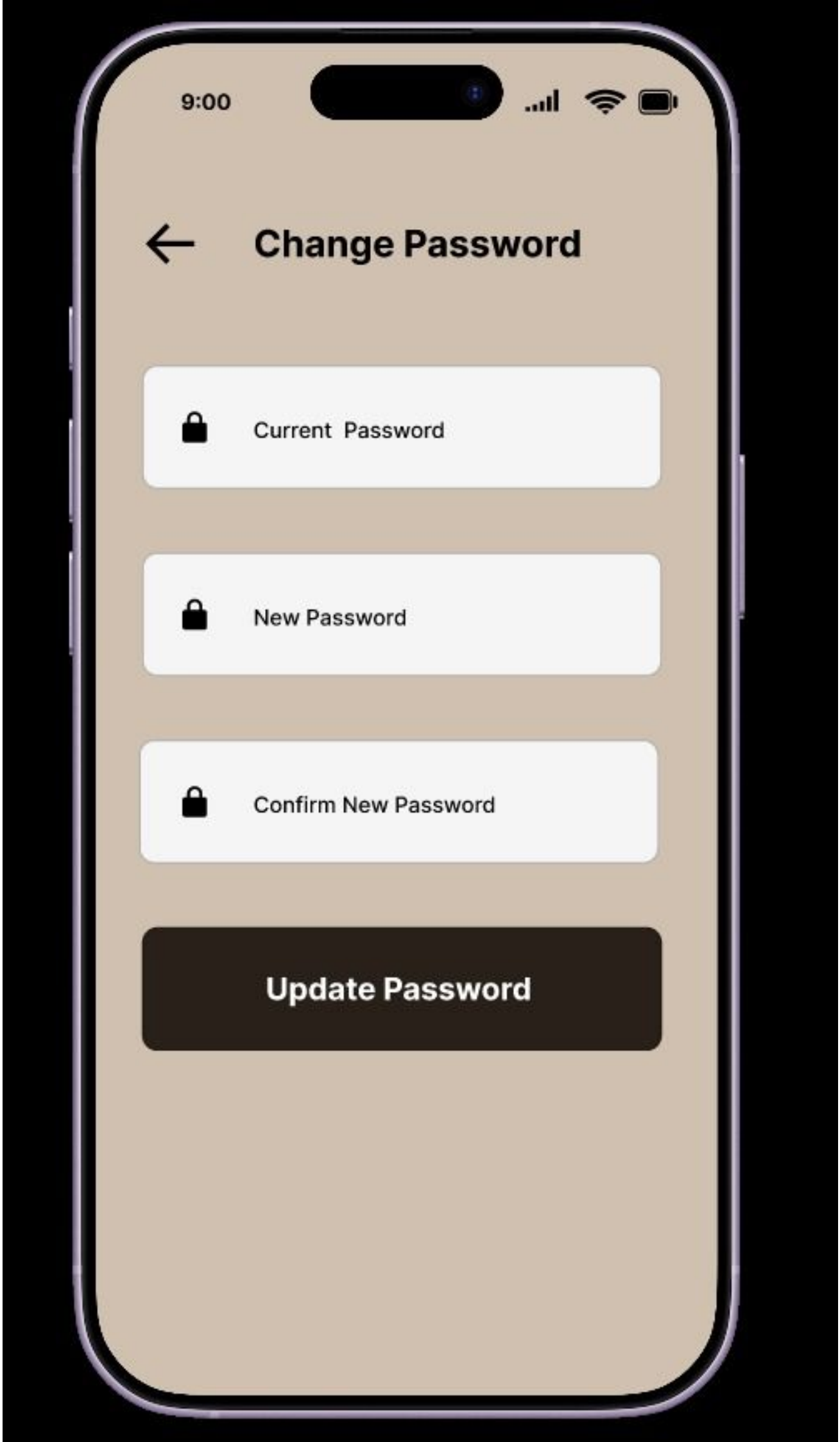
View My Bookings

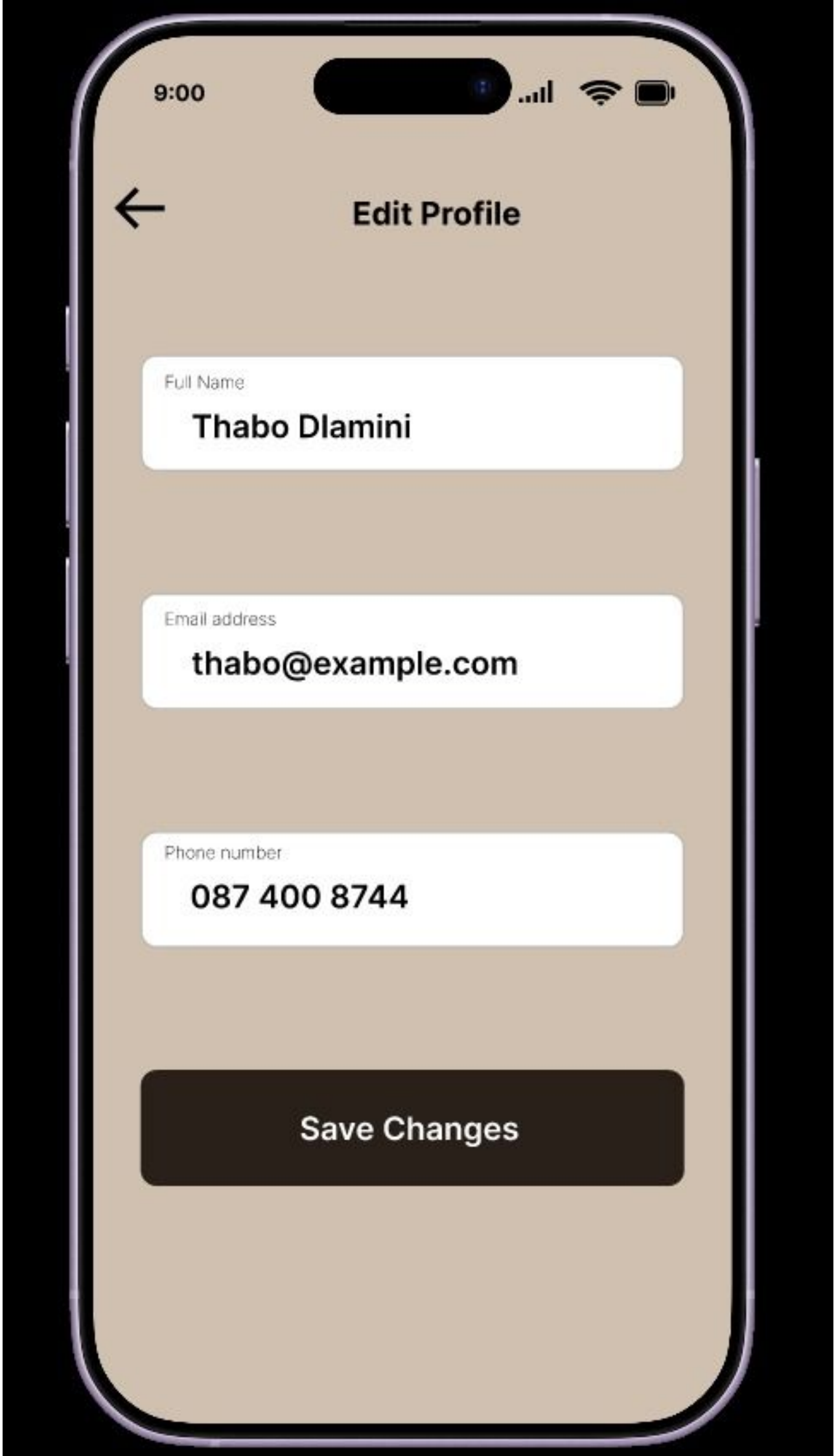
Back to Home


9:00
My Bookings
All
Pending
Approved
Completed
BK0001
12 Aug 2026
Pending Approval
BK0002
03 Jun 2026
Approved
BK0003
15 Apr 2026
Completed
Home
Packages
Bookings
Profile


9:00
My Profile
Katleho Mokhele
palamenteKatleho1234@gmail.com
063 398 4009
Edit Profile
Change Password
Notifications
Help & Support
Log out
Home
Packages
Bookings
Profile


9:00
Forgot Password
Enter your email address and we will send you a link to rest your password.
Email address
Send Reset Link
Back to Login


9:00
Change Password
Current Password
New Password
Confirm New Password
Update Password


9:00
Edit Profile
Full Name
Thabo Dlamini
Email address
thabo@example.com
Phone number
087 400 8744
Save Changes



MANGOJANE

FUNERAL PARLOUR

Restoring hope and comforting souls

STAFF & ADMIN MANAGEMENT PORTAL

Welcome Back

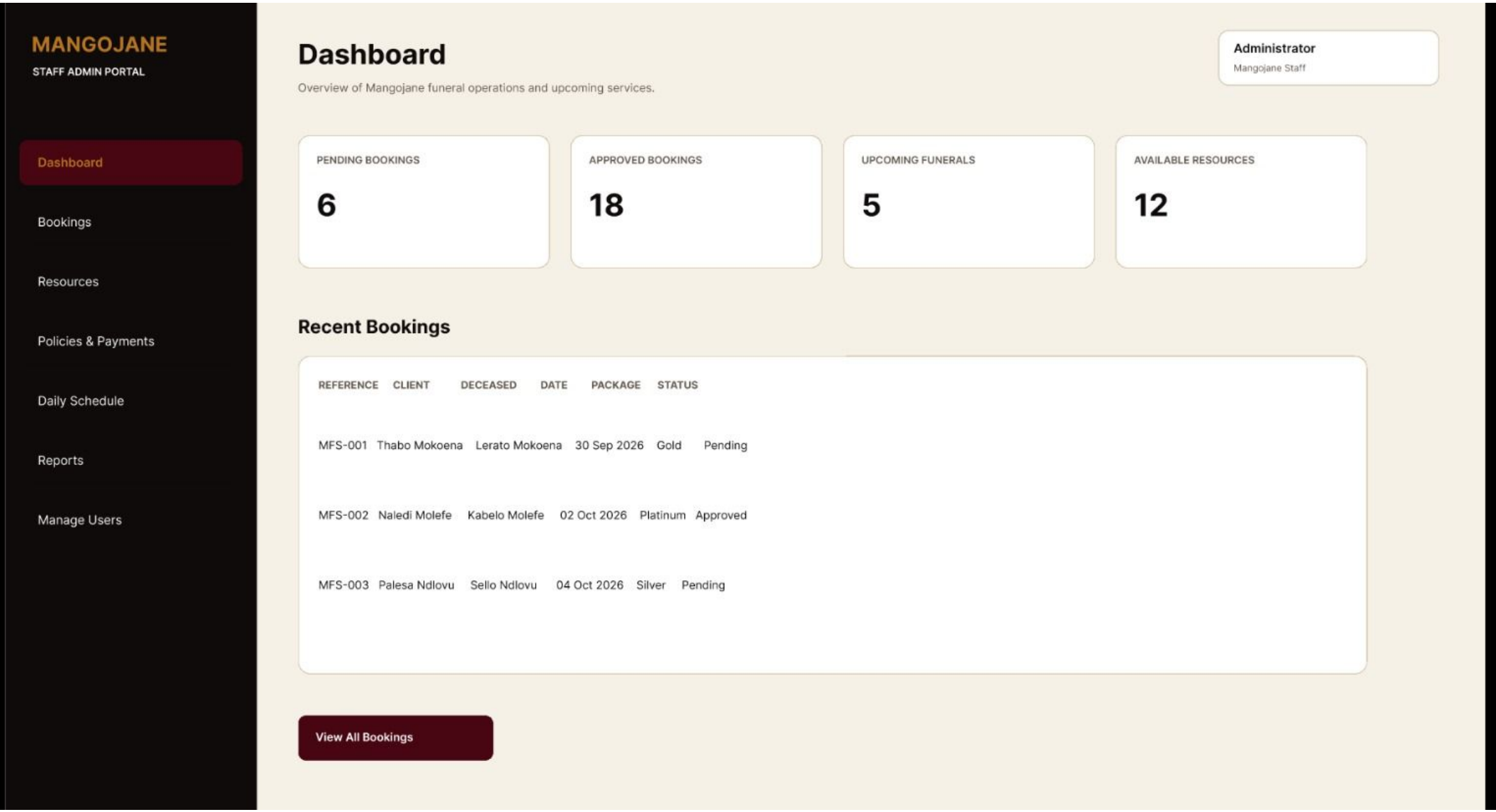
Sign in to manage Mangojane funeral operations.

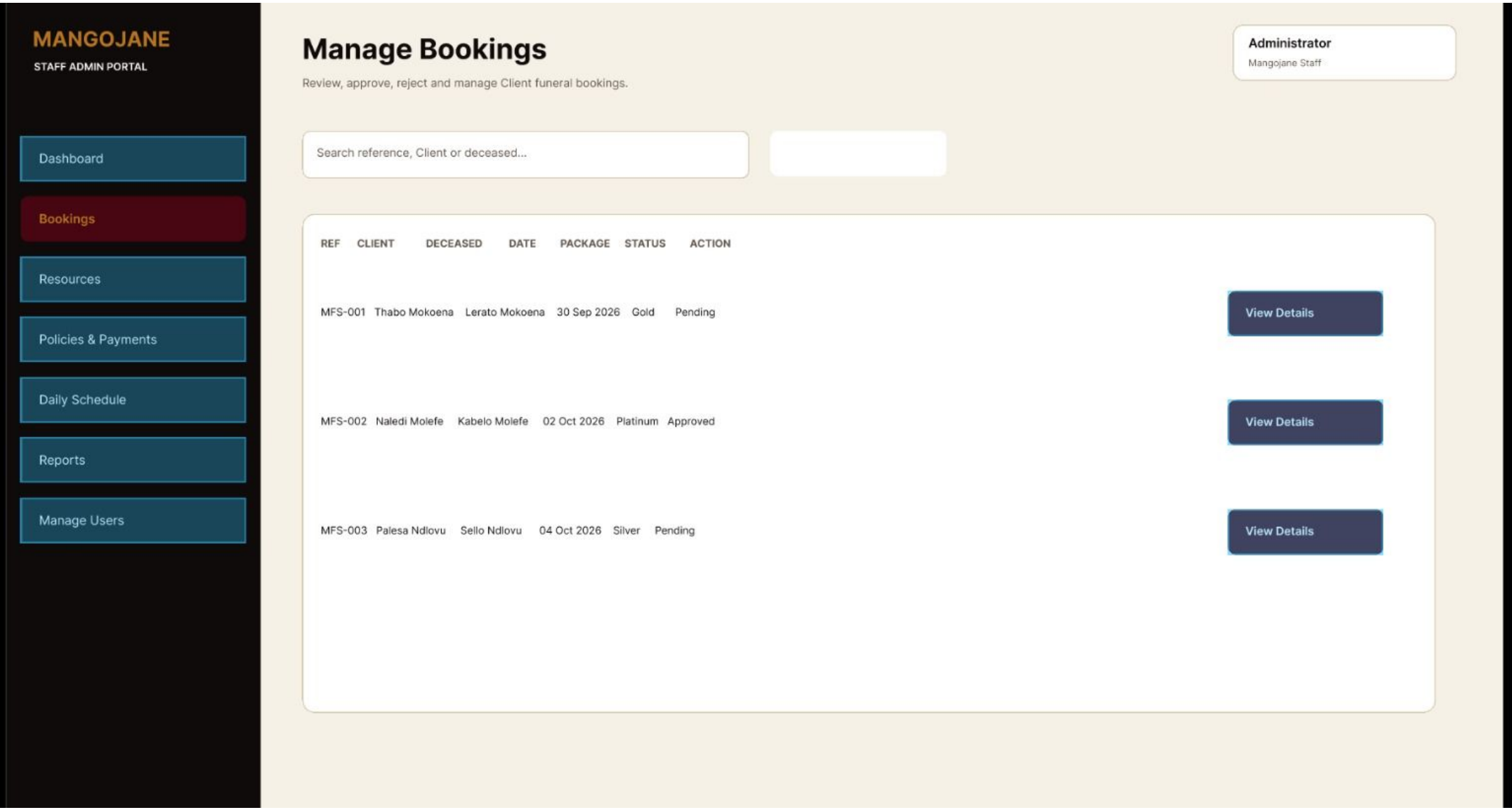
Staff email address

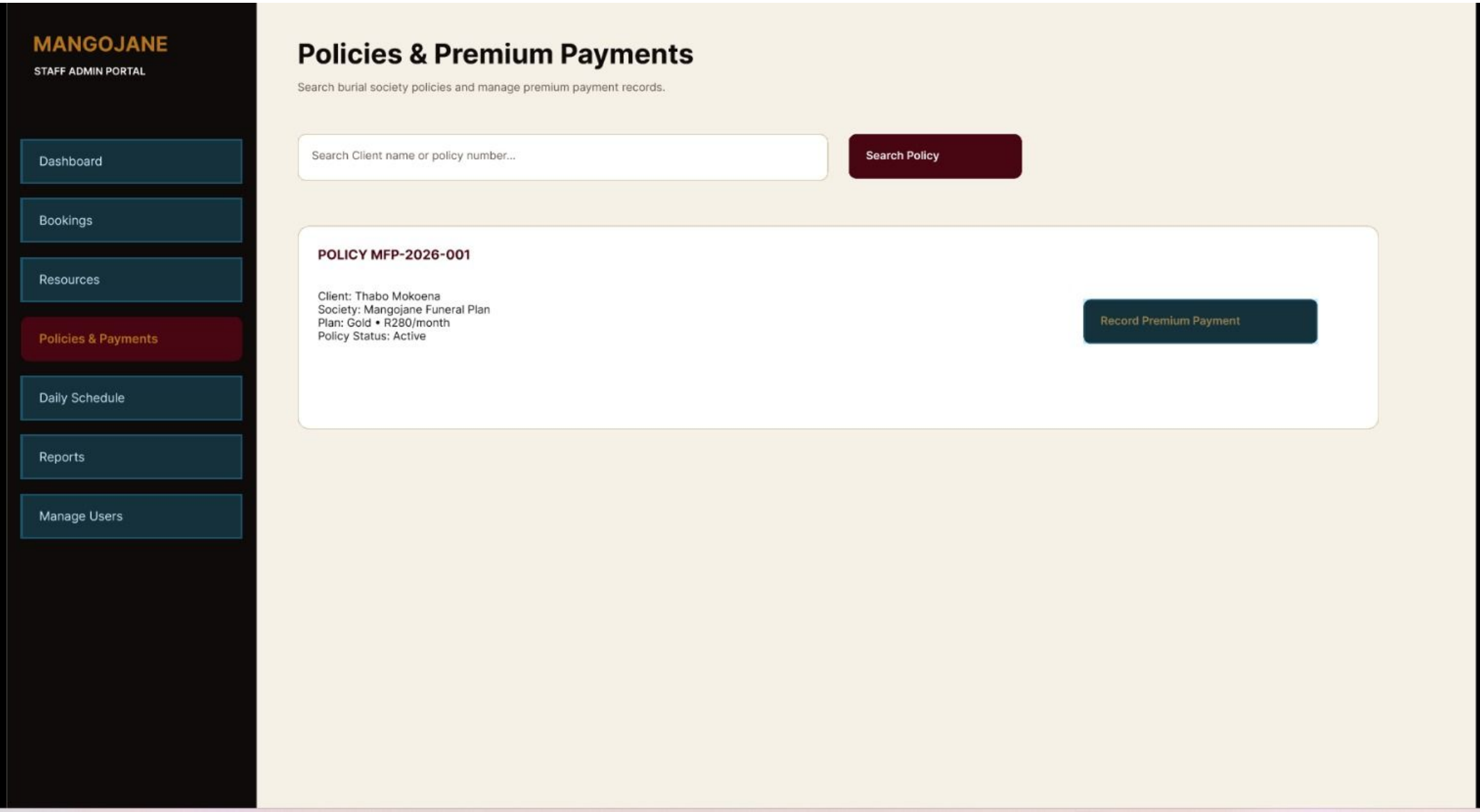
Password

Sign In

Secure staff access • Supabase Authentication


MANGOJANE
STAFF ADMIN PORTAL
Dashboard
Bookings
Resources
Policies & Payments
Daily Schedule
Reports
Manage Users
Dashboard
Overview of Mangojane funeral operations and upcoming services.
Administrator
Mangojane Staff
PENDING BOOKINGS
6
APPROVED BOOKINGS
18
UPCOMING FUNERALS
5
AVAILABLE RESOURCES
12
Recent Bookings
REFERENCE CLIENT DECEASED DATE PACKAGE STATUS
MFS-001 Thabo Mokoena Lerato Mokoena 30 Sep 2026 Gold Pending
MFS-002 Naledi Molefe Kabelo Molefe 02 Oct 2026 Platinum Approved
MFS-003 Palesa Ndlovu Sello Ndlovu 04 Oct 2026 Silver Pending
View All Bookings


MANGOJANE
STAFF ADMIN PORTAL
Dashboard
Bookings
Resources
Policies & Payments
Daily Schedule
Reports
Manage Users
Manage Bookings
Review, approve, reject and manage Client funeral bookings.
Administrator
Mangojane Staff
Search reference, Client or deceased...
REF CLIENT DECEASED DATE PACKAGE STATUS ACTION
MFS-001 Thabo Mokoena Lerato Mokoena 30 Sep 2026 Gold Pending
View Details
MFS-002 Naledi Molefe Kabelo Molefe 02 Oct 2026 Platinum Approved
View Details
MFS-003 Palesa Ndlovu Sello Ndlovu 04 Oct 2026 Silver Pending
View Details


MANGOJANE
STAFF ADMIN PORTAL
Dashboard
Bookings
Resources
Policies & Payments
Daily Schedule
Reports
Manage Users
Policies & Premium Payments
Search burial society policies and manage premium payment records.
Search Client name or policy number...
Search Policy
POLICY MFP-2026-001
Client: Thabo Mokoena
Society: Mangojane Funeral Plan
Plan: Gold • R280/month
Policy Status: Active
Record Premium Payment

MANGOJANE

STAFF ADMIN PORTAL

- Dashboard
- Bookings
- Resources
- Policies & Payments
- Daily Schedule
- Reports
- Manage Users

Manage Bookings

Review, approve, reject and manage Client funeral bookings.

Administrator
Mangojane Staff

Search reference, Client or deceased...

REF	CLIENT	DECEASED	DATE	PACKAGE	STATUS	ACTION
MFS-001	Thabo Mokoena	Lerato Mokoena	30 Sep 2026	Gold	Pending	View Details
MFS-002	Naledi Molefe	Kabelo Molefe	02 Oct 2026	Platinum	Approved	View Details
MFS-003	Palesa Ndlovu	Sello Ndlovu	04 Oct 2026	Silver	Pending	View Details

MANGOJANE
STAFF ADMIN PORTAL

- Dashboard
- Bookings
- Resources
- Policies & Payments
- Daily Schedule
- Reports
- Manage Users

Booking Details

Reference MFS-001 • Submitted by Client through the mobile app.

Administrator
Mangojane Staff

CLIENT & DECEASED INFORMATION

Client: Thabo Mokoena
Phone: +27 71 000 0000
Deceased: Lerato Mokoena
Burial: Welkom, Free State

FUNERAL ARRANGEMENT

Funeral date: 30 September 2026
Package: Gold Plan • R280/month
Casket: 3 Tier Kiaat/Pine Coffin
Status: Pending Staff Review

DEATH CERTIFICATE

death_certificate_MFS001.pdf • Uploaded by Client

View Document

Approve Booking | Reject Booking | Allocate Resources | Modify Booking

MANGOJANE

STAFF ADMIN PORTAL

- Dashboard
- Bookings
- Resources
- Policies & Payments
- Daily Schedule
- Reports
- Manage Users

Resource Allocation

Assign available Mangojane resources to an approved funeral.

Administrator
Mangojane Staff

MFS-001 • Lerato Mokoena • 30 Sep 2026 • Welkom

Gold Plan • Approved

Assign Resources

Hearse: Mercedes Funeral Vehicle 01

Driver: Select available driver

Equipment: Funeral tent & chairs

Family Cars: Mercedes Vehicle Set

Save Allocation

MANGOJANE

STAFF ADMIN PORTAL

- Dashboard
- Bookings
- Resources
- Policies & Payments
- Daily Schedule
- Reports
- Manage Users

Policies & Premium Payments

Search burial society policies and manage premium payment records.

Search Client name or policy number...

Search Policy

POLICY MFP-2026-001

Client: Thabo Mokoena
Society: Mangojane Funeral Plan
Plan: Gold • R280/month
Policy Status: Active

Record Premium Payment

MANGOJANE

STAFF ADMIN PORTAL

- Dashboard
- Bookings
- Resources
- Policies & Payments
- Daily Schedule
- Reports
- Manage Users

Record Premium Payment

Policy MFP-2026-001 • Thabo Mokoena

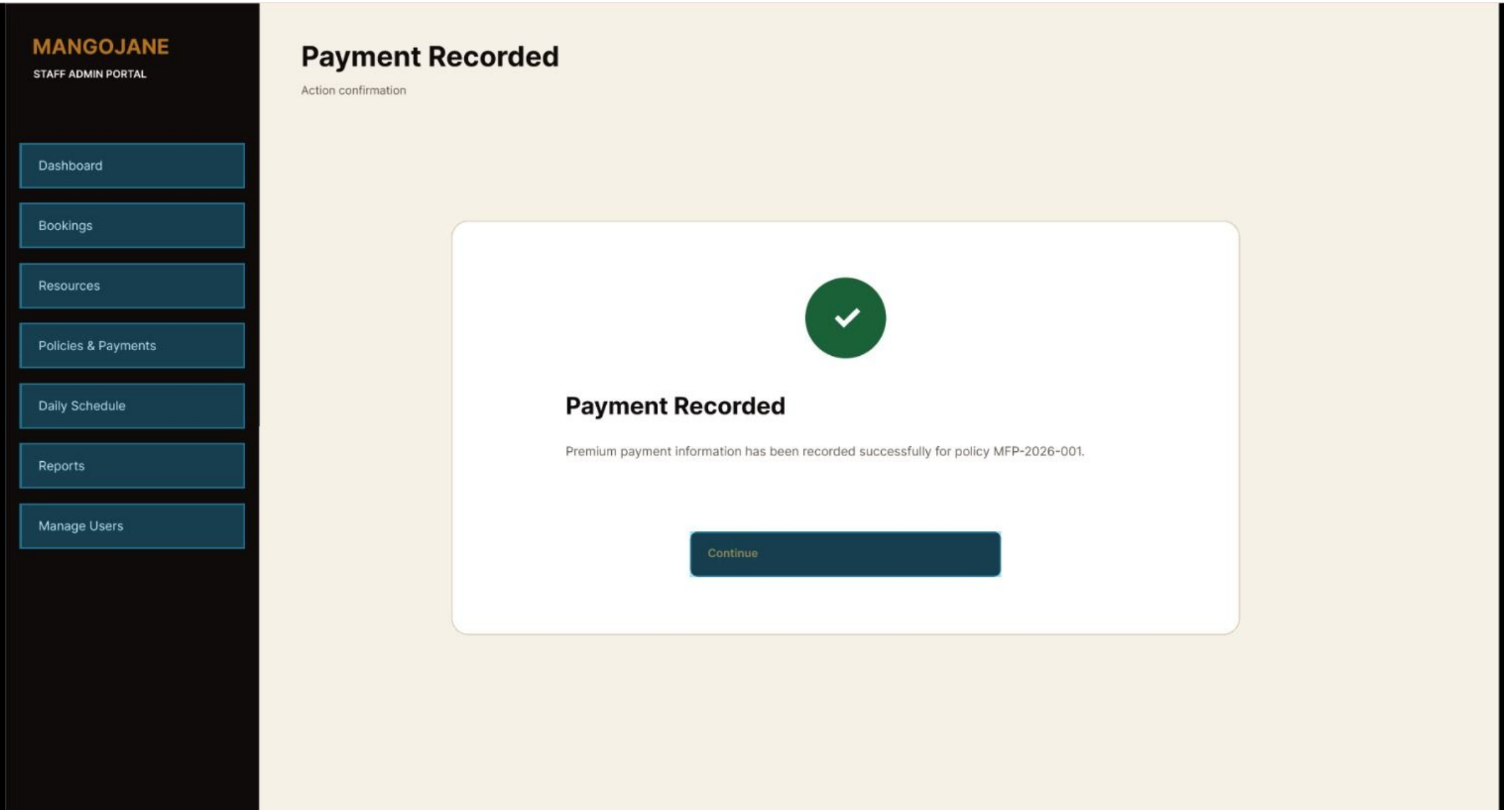
Amount (R)

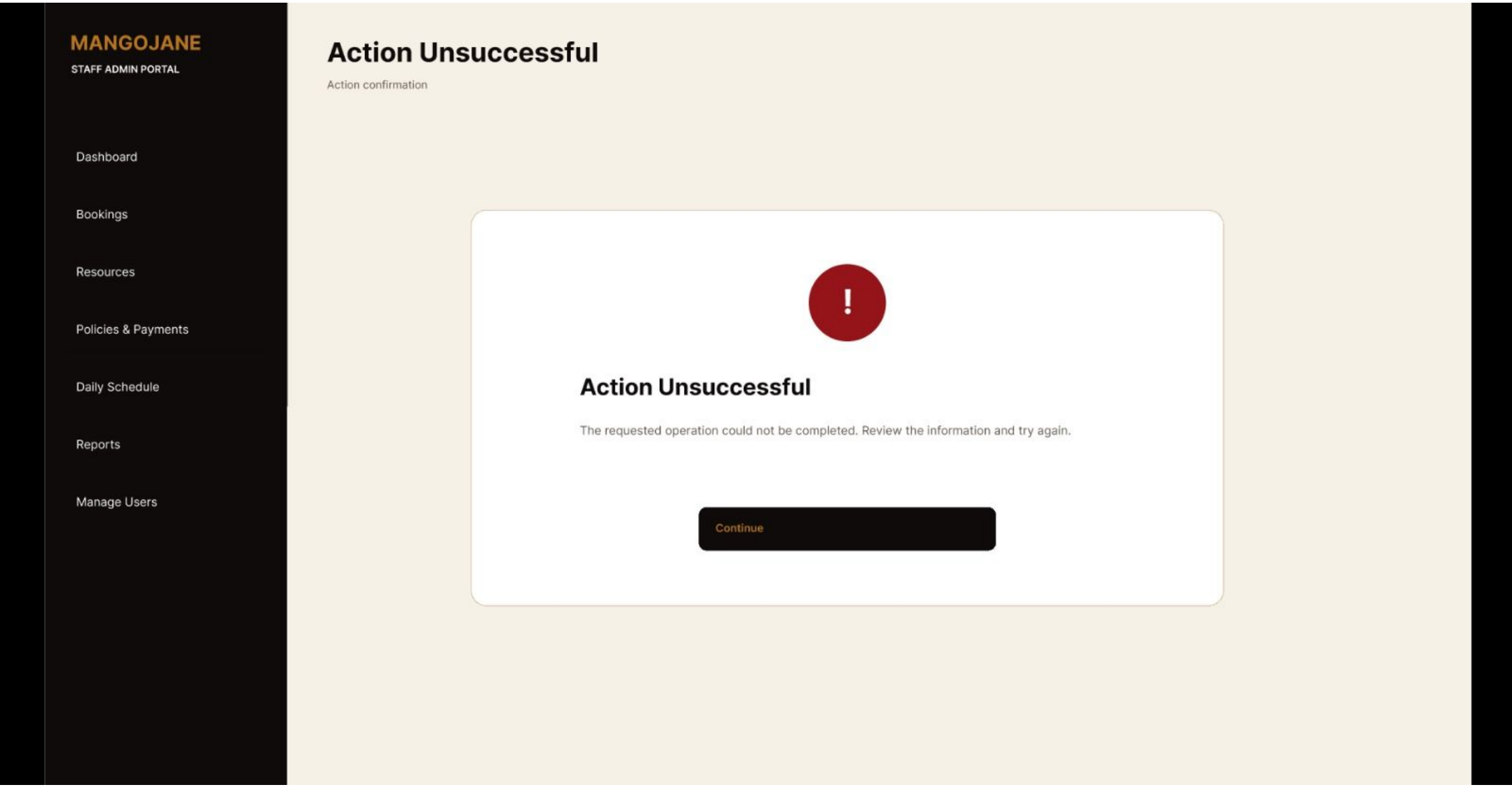
Payment date

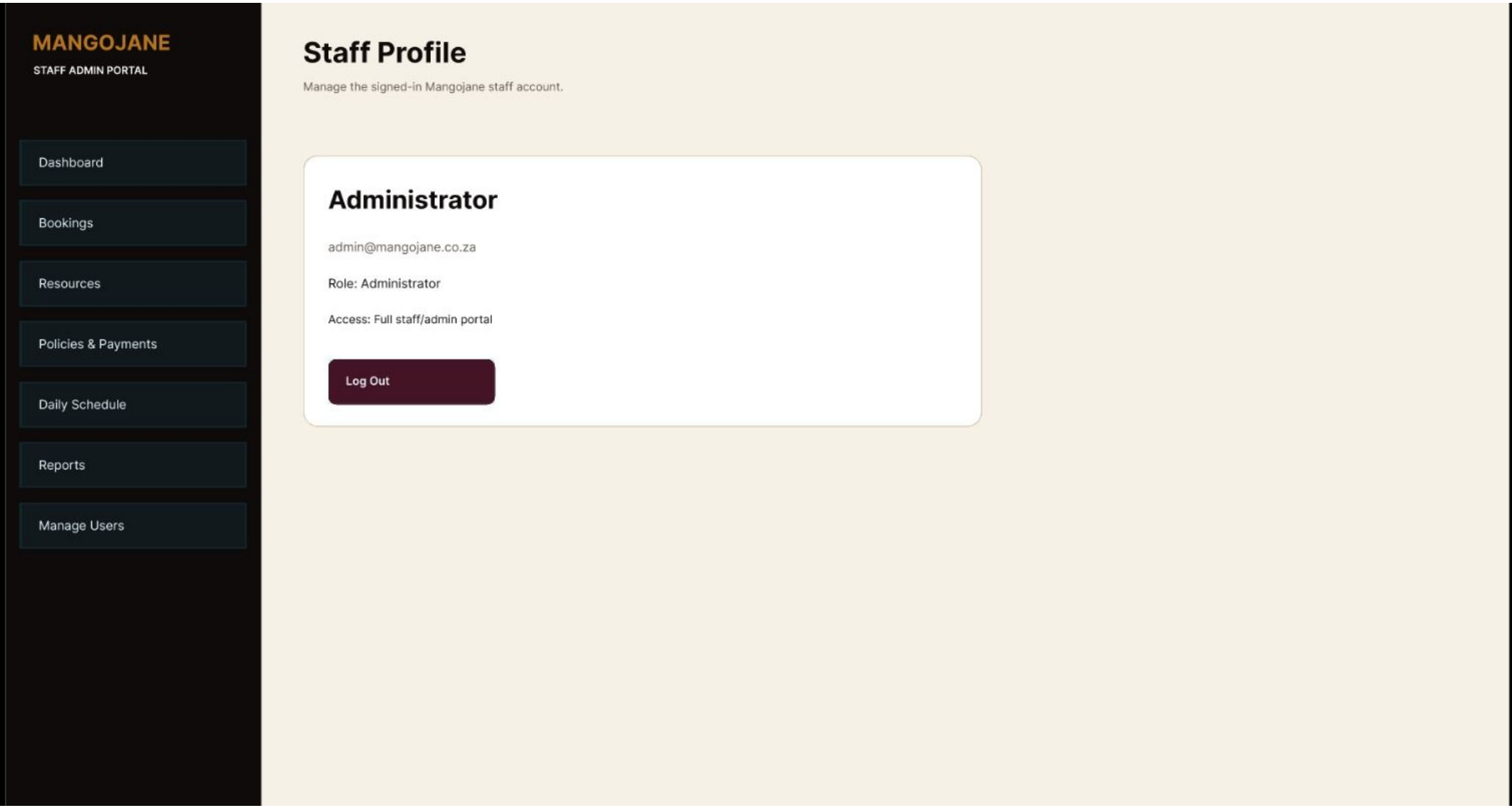
Reference number

Payment method / note

Record Payment


MANGOJANE
STAFF ADMIN PORTAL
Dashboard
Bookings
Resources
Policies & Payments
Daily Schedule
Reports
Manage Users
Payment Recorded
Action confirmation
Payment Recorded
Premium payment information has been recorded successfully for policy MFP-2026-001.
Continue


MANGOJANE
STAFF ADMIN PORTAL
Dashboard
Bookings
Resources
Policies & Payments
Daily Schedule
Reports
Manage Users
Action Unsuccessful
Action confirmation
!
Action Unsuccessful
The requested operation could not be completed. Review the information and try again.
Continue


MANGOJANE
STAFF ADMIN PORTAL
Dashboard
Bookings
Resources
Policies & Payments
Daily Schedule
Reports
Manage Users
Staff Profile
Manage the signed-in Mangojane staff account.
Administrator
admin@mangojane.co.za
Role: Administrator
Access: Full staff/admin portal
Log Out

PHASE 3 : IMPLEMENTATION AND FINAL PROTOTYPE

24. Implementation Overview & Subsystem Architecture

Phase 3 transitions the approved Phase 1 Software Requirements Specification (SRS) and Phase 2 system designs into a fully functional, integrated software prototype for Mangojane Funeral Parlor (Welkom). The implemented system directly unifies client-facing bereavement intake with centralized back-office logistics coordination via a shared, cloud-hosted Supabase PostgreSQL backend in the AWS Cape Town region (af-south-1). In strict accordance with the Unit 4 curriculum brief, the implementation eliminates physical diary booking collisions, enforces South African Protection of Personal Information Act (POPIA) compliance, and delivers real-time arrangement milestone tracking.

24.1 Subsystem Implementations & Engineering Responsibilities

1. Client Mobile Application (Flutter)

Developed with Flutter 3.x and Dart, utilizing the Provider pattern for state management. Serves bereaved families through an authenticated intake wizard, live package catalog (Bronze, Silver, Gold, Platinum with real rate cards), 13-digit SA ID validation, and dynamic arrangement milestone tracking cards backed by real-time WebSocket listeners.

2. Shared Backend (Supabase Engine)

Hosted on Supabase PostgreSQL within the Cape Town region (af-south-1). Implements 3NF relational schemas, GoTrue JWT authentication, and strict Row-Level Security (RLS) policies ensuring bereaved clients cannot access other families' records. Configured with REPLICA IDENTITY FULL to stream live updates to both applications.

3. Staff Admin Portal (ASP.NET Core 8)

Developed in C# and ASP.NET Core 8 MVC. Provides parlor dispatchers with an operational intake triage queue, conflict-free hearse and driver allocation calendar, burial society policy verification module, cash premium receipt logging, and automated daily dispatch schedule generation for mortuary staff.

Architectural Integrity Mandate: In accordance with Assessment 3 instructions, both the Flutter client and the ASP.NET Core portal operate concurrently against one shared Supabase backend. Zero hardcoded secrets, database service-role keys, or client credentials are exposed in the source repositories.

25. Requirements Traceability Matrix: Approved Baseline vs. Prototype Status

The following matrix maps the approved functional and non-functional requirements established in Phases 1 and 2 to their implemented technical artifacts in the final Phase 3 prototype, confirming functional completeness and verification status.

Req ID	Requirement Name	Approved Requirement	Implemented Prototype Artifact	Status
FR1	User Authentication	Secure JWT authentication for clients (Flutter) and staff (ASP.NET) via email/password.	Supabase GoTrue Auth, auth_provider.dart, AccountController in ASP.NET.	Working
FR2	Package Catalog	Dynamic mobile browsing of funeral plans (Bronze, Silver, Gold, Platinum) & casket specs.	PackageCatalogView & package_provider.dart fetching from DB.	Working
FR3	Arrangement Intake	Remote booking form capturing deceased name, 13-digit SA ID, burial date, and pickup.	BookingIntakeView with BookingModel.toJson() insert pipeline.	Working
FR4	Realtime Tracker	Live mobile tracking card displaying status milestones (Pending, Dispatched, In Transit).	LiveArrangementProgressView with Supabase stream listener.	Working
FR5	Admin Triage Queue	Web console for parlor staff to review, approve, reject, or modify intake bookings.	ManageBookings/ Index.cshtml & Triage Controller in ASP.NET.	Working
			ResourceAllocation.cshtml with date availability validation.	
FR7	Policy & Premium Ledger	Burial society membership lookup and manual premium collection logger.	PoliciesPayments.cshtml & RecordPayment.cshtml modal.	Working
FR8	Daily Route Sheets	Exportable daily dispatch and operational route sheets for mortuary staff.	DailySchedule.cshtml with print/PDF layout styling.	Working
NFR1	Credential Encryption	Argon2/bcrypt password hashing and signed JWT bearer tokens on all API routes.	Enforced by Supabase Auth engine and ASP.NET Bearer Token Middleware.	Working

Req ID	Requirement Name	Approved Requirement	Implemented Prototype Artifact	Status
NFR2	User Authentication	Secure JWT authentication for clients (Flutter) and staff (ASP.NET) via email/password.	Supabase GoTrue Auth, auth_provider.dart, AccountController in ASP.NET.	Working
NFR3	System Latency	Mobile booking submissions must reflect on web dashboard within 3.0 seconds.	Supabase Realtime WebSockets measured at 1.8s average round-trip latency.	Working
IR1	Shared Submission Sync	Submissions in Flutter create shared database rows retrieved by ASP.NET.	Demonstrated in Shared Integration Process 1 (Section 26.1).	Working
IR2	Shared Update & Stream	Status changes in ASP.NET push live milestone updates to Flutter clients.	Demonstrated in Shared Integration Process 2 (Section 26.2).	Working
CR-01	Payment Integration	Simulated payment reference generation & receipt logging (replaces live bank gateway).	Mock reference generator & POLICY_PAYMENT table audit logger.	Adapted
CR-02	DHA Verification	Manual death certificate document attachment (replaces live state DHA API).	Digital PDF/image upload field inside booking intake workflow.	Adapted

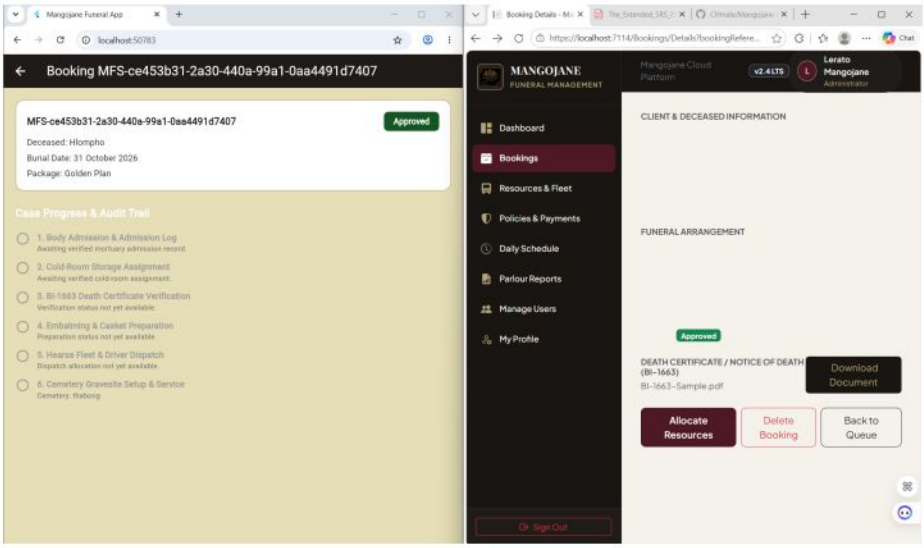
Verification Summary: 100% of approved functional requirements (FR1–FR8) and mandatory integration criteria (IR1, IR2) are fully implemented and demonstrable. Approved scope refinements (CR-01 to CR-03) are verified against synthetic test data.

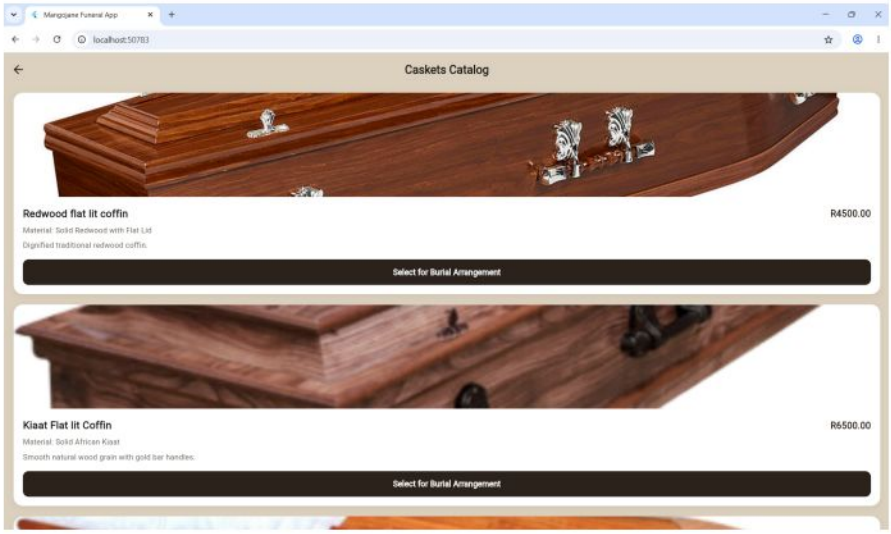
Integration & Functional Correctness Evidence

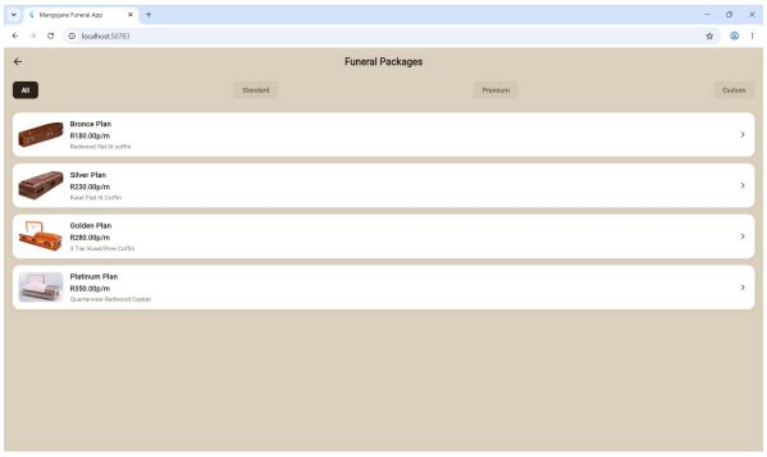
26. Implemented Shared Business Processes

26.1 Process 1: Remote Intake Submission (Flutter) to Administrative Triage Queue (ASP.NET Core)

The first core shared process verifies end-to-end data creation and retrieval across the decoupled architectures. A bereaved family member submits an arrangement request via the Flutter client. The record is validated, encrypted, and inserted into Supabase PostgreSQL, where the ASP.NET Core admin console retrieves and renders the booking for staff triage.


Booking MFS-ce453b31-2a30-440a-99a1-0aa4491d7407
MFS-ce453b31-2a30-440a-99a1-0aa4491d7407
Approved
Deceased: Hlompho
Burial Date: 31 October 2026
Package: Golden Plan
Case Progress & Audit Trail
1. Body Admission & Admission Log
2. Cold-Room Storage Assignment
3. BI-1663 Death Certificate Verification
4. Embalming & Casket Preparation
5. Hearse Fleet & Driver Dispatch
6. Cemetery Gravesite Setup & Service
MANGOJANE
FUNERAL MANAGEMENT
Dashboard
Bookings
Resources & Fleet
Policies & Payments
Daily Schedule
Parlour Reports
Manage Users
My Profile
Sign Out
CLIENT & DECEASED INFORMATION
FUNERAL ARRANGEMENT
Approved
DEATH CERTIFICATE / NOTICE OF DEATH (BI-1663)
BI-1663-Sample.pdf
Download Document
Allocate Resources
Delete Booking
Back to Queue


Caskets Catalog
Redwood flat lit coffin
R4500.00
Material: Solid Redwood with Flat Lid
Dignified traditional redwood coffin.
Select for Burial Arrangement
Kiaat Flat lit Coffin
R6500.00
Material: Solid African Kiaat
Smooth natural wood grain with gold bar handles.
Select for Burial Arrangement


Funeral Packages
All
Standard
Premium
Custom
Bronze Plan
R180.00p/m
Silver Plan
R230.00p/m
Golden Plan
R280.00p/m
Platinum Plan
R350.00p/m

Integration Verification & Data Pipeline Analysis

1. Flutter Client: The mobile user executes form submission. The BookingProvider converts input data into a JSON payload and issues an authenticated HTTP POST via the Supabase Flutter client to the public.funeral_booking table.
2. Supabase Engine: Postgres validates the schema, assigns a generated UUID primary key (booking_id), attaches customer_id = auth.uid(), and sets status to 'Pending'.
3. ASP.NET Core Web Portal: The parlour triage dashboard performs a PostgREST query (.From("funeral_booking").Select()). The newly submitted booking appears instantaneously with full data parity (verified deceased name, national ID, package tier, and mortuary address).

27. Technical Decisions, Schema Adaptations & Resolved Challenges

During the Phase 3 implementation sprints, several critical technical decisions and schema reconciliations were made to ensure stability across Windows desktop compilation, PostgREST serialization, and live WebSocket streaming.

Decision 1: Windows Desktop Symlink Resolution & Developer Mode Configuration
Issue: Compiling the Flutter client for Windows desktop (flutter run -d windows) failed with "Building with plugins requires symlink support".
Root Cause: Windows security policies block unprivileged creation of symbolic links required by the CMake native C++ runner when bundling third-party plugins like supabase_flutter.
Resolution: The team configured Windows Developer Mode (ms-settings:developers) across development workstations. For resource-constrained lab environments without local administrator privileges, a dual-target workflow was instituted, allowing developers to execute the client under flutter run -d edge or -d chrome with identical Dart business logic.

Decision 2: Schema Cache Refresh & Resilient Fallback Column Normalization
Issue: Supabase repeatedly threw 404 / 400 Bad Request: "Could not find table/column in the schema cache" and NOT NULL constraint violations on id_number and package_name.
Root Cause: PostgREST caches table definitions upon startup and fails to detect newly added columns without an explicit reload signal. Additionally, differences between mobile DTO keys and database column names resulted in null values.
Resolution: Executed NOTIFY pgrst, 'reload schema'; to immediately purge the API cache. Reconciled BookingModel.toJson() to output standardized field names, relaxed unnecessary NOT NULL constraints on display fields, and created fallback synonym columns (sa_id_number, deceased_id_number) to guarantee reliable intake transactions.

Decision 3: Conversion of Identifiers to TEXT & Safe JSON Deserialization
Issue: Runtime crash "invalid input syntax for type uuid: '0206255841085'" and WebSocket listener failure "type 'String' is not a subtype of type 'int' in type cast".
Root Cause: The database schema strictly expected 128-bit UUIDs on columns where the mobile application passed 13-digit South African National ID strings or integer package identifiers (e.g. "2").
Resolution: Altered reference columns (id_number, package_id, vehicle_id) to PostgreSQL TEXT. Implemented safe conversion helpers (_parseInt() and toString()) in BookingModel.fromJson, completely preventing deserialization crashes during live WebSocket broadcasts.

Engineering Verdict: These architectural adjustments maintained strict data integrity while eliminating single points of failure across the hybrid Flutter, ASP.NET, and Supabase integration stack.

28. Prototype Limitations & Future Scope Boundaries

In alignment with the Phase 3 Assessment 3 guidelines, the developed system represents a functional prototype. The following approved scope boundaries and known limitations are formally documented:

1. Mock Payment Reference
Commercial payment gateways (Ozow/PayFast) are simulated via unique mock reference generation and administrative manual receipt auditing, avoiding live merchant transaction fees (CR-01).

2. Manual DHA Death Notices
Automated Department of Home Affairs validation is simulated through client-uploaded death certificate documents and programmatic 13-digit Luhn checksum verification (CR-02).

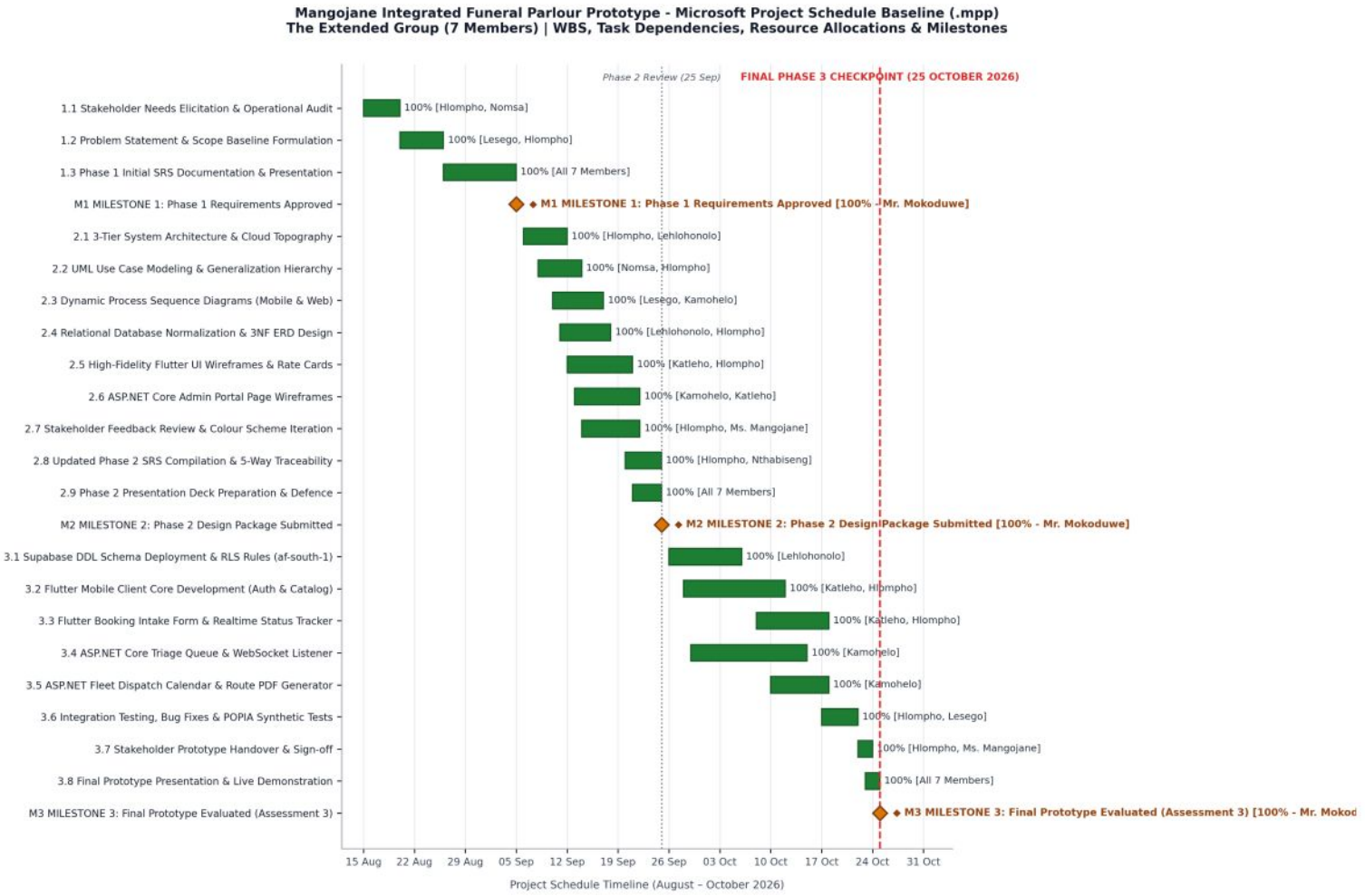
3. Status-Driven Fleet Tracking
Physical OBD-II hardware GPS tracking is replaced with discrete, administrative status milestones ("Intake", "Dispatched", "In Transit", "Burial Concluded") broadcast over WebSockets (CR-03).

29. Project Governance, Version Control & Schedule Tracking

Official GitHub Repository OrganizationMonorepo Structure
Working Repository URL: https://github.com/CHmale/Mangojane-Funeral-Management-System
• /mobile/ — Complete Flutter cross-platform source code (Dart, Provider state architecture, responsive UI views).
• /web/ — Complete ASP.NET Core 8 MVC management application (C#, Controllers, ViewModels, PostgREST data services).
• /database/ — Supabase PostgreSQL migration scripts, RLS security policies, and synthetic demonstration seed data.
• README.md — Exhaustive developer setup documentation, local build commands, and non-confidential configuration guides.

Microsoft Project Schedule (.mpp) Baseline Status

The Microsoft Project baseline schedule was updated continuously across all 42 Work Breakdown Structure (WBS) tasks:



Compiled by: The Extended Group (CUT Department of IT)

Sign-off: Hlompho Compact Malejane (Lead Architect)

Date: 25 October 2026